MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

CONVERTER, S/D, KEY CODE JAP

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic S/D converter module.

1.2 Classification. This specification covers two classes of modules.

Class 1 and class II as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence

3 REQUIREMENTS

3.1 General requirements The module shall comply with the requirements of MIL-M-28787 except as modified herein

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Naval Electronic Systems Command, ATTN ELEX 8111, Department of the Navy, Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 1.06 failures per million operating hours for class I or 5.36 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1399. When a class II module is used in a class I environment, failure rate for class I shall be used for reliability prediction (see 6.3).

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I. Class I modules shall conform to the electrical requirements of class I only, and class II modules shall conform to the electrical requirements of class I and class II (see 6.3).

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be JAP.

3.5.2 Size. The module size shall be 1A, format A.

3.5.3 Weight. The module shall have a maximum weight of 45 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/345-1 for class I and M28787/345-2 for class II, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.7).

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

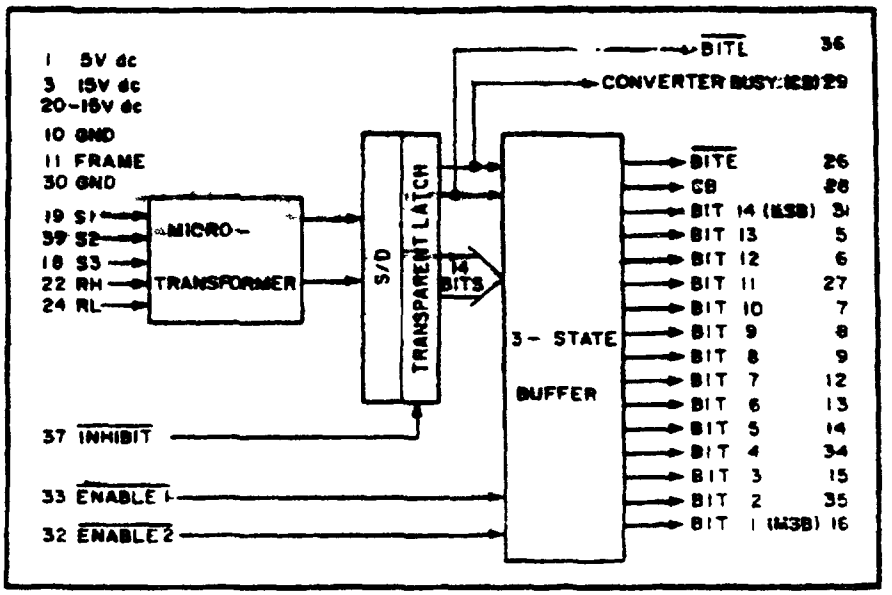
4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

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NOTE Pins 2, 4, 17, 21, 23, 25, 38, and 40 are not used.

FIGURE 1 JAF module function diagram, S/D converter

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TABLE I Module electrical requirements

Test condition	Parameter	Ref test table	Test type 1/ 3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A1 A2	Continuity	II	X	0	1	0	1	0	1	0	1	0	1	-
B1-B9	Isolation													
	Initial	III	X	10	"	10	"	10	"	10	"	10	"	MΩ
	End-of-life	III	X	1		1		1	"	1	"	1	"	MΩ
C	Insulation resistance													
	Initial	III	Q	10	"	10	"	10	"	10	"	10	"	MΩ
	End-of-life	III	Q	1	"	1		1	"	1	"	1	"	MΩ
D	Common mode voltage	IV	Q	←					Survival					→
E1-E6	Power supply sequencing	V	Q	←					Survival					→
F1	Power supply current	VI	X	0.1	30.0	0.1	30.0	0.1	30.0	0.1	30.0	0.1	30.0	mA
F2	Power supply current	VI	X	0.01	30.00	-0.01	30.00	-0.01	30.00	-0.01	30.00	-0.01	30.00	mA
F3	Power supply current	VI	X	0.1	130.0	0.1	130.0	0.1	130.0	0.1	130.0	0.1	130.0	mA
G1	Power supply capacitance	VII	X	1.82	3.38	1.82	3.38	1.82	3.38	1.82	3.38	1.82	3.38	µF
G2	Power supply capacitance	VII	X	1.96	3.64	1.96	3.64	1.96	3.64	1.96	3.64	1.96	3.64	µF
H1-H17	Static angular accuracy, V_{normal}													
	Initial	VIII	X	-4	4	-4	4	-4	4	-4	4	-4	4	LSB
	End-of-life	VIII	X	-5	5	-5	5	-5	5	-5	5	-5	5	LSB
J1-J17	Static angular accuracy, V_{low}													
	Initial	IX	S	-4	4	-4	4	-4	4	-4	4	-4	4	LSB
	End-of-life	IX	S	-5	5	-5	5	-5	5	-5	5	-5	5	LSB
K1 K17	Static angular accuracy, V_{high}													
	Initial	X	S	-4	4	-4	4	-4	4	-4	4	-4	4	LSB
	End-of-life	X	S	-5	5	-5	5	-5	5	-5	5	-5	5	LSB

See footnotes at end of table

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TABLE I Module electrical requirements Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
L	Sensitivity check	XI	S	0 010	0 033	0.010	0 033	0 010	0 033	0.010	0.033	0 010	0.033	Degree
M1-M14	Digital output voltage (V_{OH})	XII	X	2 4	5 1	2 4	5.1	2.4	5 1	2 4	5.1	2.4	5.1	V
N1-N14	Digital output voltage (V_{OL})	XIII	X	0	400	0	400	0	400	0	400	0	400	mV
P1-P3	Digital input breakdown current	XIV	S	0	40	0	40	0	40	0	40	0	40	uA
Q1-Q3	Digital input current	XV	S	-1 6	0.0	-1 6	0.0	-1 6	0.0	-1.6	0.0	-1 6	0.0	mA
R1-R16	Three-state leakage current, high level	XVI	X	-1	20	1	20	-1	20	-1	20	-1	20	uA
S1 S16	Three state leakage current, low level	XVII	X	1	20	1	-20	1	-20	1	-20	1	-20	uA
T	Delay time	XVIII	Q	0	3	0	3	0	3	0	3	0	3	us
U1,U2	Delay time	XVIII	Q	0	150	0	150	0	150	0	150	0	150	ns
V1,V2	Tracking rate check	XIX	Q	10	15	10	15	10	15	10	15	10	15	rps
W	Settling time	XX	Q	0	200	0	200	0	200	0	200	0	200	ms
X	Noise sensitivity	XXI	Q	-1	1	-1	1	-1	1	-1	1	-1	1	LSB
Y1	Converter busy, direct (V_H)	XXII	X	2 4	5 1	2 4	5 1	2 4	5.1	2.4	5 1	2.4	5.1	V
Y2	Converter busy, direct (V_L)	XXII	X	0	400	0	400	0	400	0	400	0	400	mV
Y3	Converter busy, direct (P_W)	XXII	X	1.5	6.0	1 5	6 0	1.5	6.0	1.5	6.0	1.5	6.0	us
Y4	Converter busy, 3-state (V_H)	XXII	X	2 4	5 1	2 4	5 1	2 4	5 1	2.4	5 1	2.4	5.1	V
Y5	Converter busy, 3-state (V_L)	XXII	X	0	400	0	400	0	400	0	400	0	400	mV
Y6	Converter busy, 3-state (P_W)	XXII	X	1.5	6.0	1 5	6 0	1 5	6.0	1.5	6.0	1.5	6.0	us

See footnotes at end of table

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TABLE I Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/ 3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
Z1-Z3	Power supply ripple rejection	XXIII	Q	1	1	-1	1	-1	1	-1	1	-1	1	LSB
AA	Continuous conversion	XXIV	Q	----	-- ------	- -- -		-----See table XXIV----------						
BB1, BB2	Power supply transient amplitude	XXV	Q	0	700	0	700	0	700	0	700	0	700	mV pt
CC	Stress	XXVI	Q	----- -- ---- --- - -- - ----------See table XXVI----------										

1/ "K" parameters shall be 100 percent tested during module acceptance tests.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C

3/ "Q" parameters shall be tested during qualification testing only

4/ Unless otherwise specified, limits are for initial and end-of-life.

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4.4.2 Order of testing. Test conditions A through CC shall be performed in any order with the following exceptions. If test conditions A, B, C, D, E, or CC are part of a sequence of tests, the order of testing shall be as follows. A, B, C, D, E, CC where applicable, followed by the remaining tests in the sequence. If at any time during quality assurance inspection testing, test conditions A, B, C, D, E, or CC are performed, they shall be followed by performance of all tests in table I which are marked with an "X" in the test type column.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through CC in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions F through S and Y shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 3. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column shall be performed using the initial limits at 25°C ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests F through S shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests A, B, and F through BB shall be performed in the order specified in 4.4.2 using the 25°C ±5°C ambient end-of-life requirements of table I.

5 PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6 NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in a class I environment, the failure rate for class I shall be used for reliability prediction (see 3.3 and 3.4).

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6.4 Module description. The following electrical parameters reflect the 0°C to 60°C and -55°C to 85°C end-of-life limits from table I.

	Class I	Class II	
Power dissipation	1.9 max	1.9 max	W
Input voltage			
Low level	0 min, 400 max	0 min, 400 max	mV
High level	2.4 min, 5.1 max	2.4 min, 5.1 max	V
Input current			
Low level (V_{IL} = 0.4V)	0.0 min, 1.6 max	0.0 min, 1.6 max	mA
High level (V_{IH} = 2.4V)	0 min, 40 max	0 min, 40 max	µA
Output drive current			
Three state low level (V_{OL} = 0.4V)	12	12	mA
Three state high level (V_{OH} = 2.4V)	400	400	µA
Direct output low level (V_{OL} = 0.4V)	1.2	1.2	mA
Direct output high level (V_{OH} = 2.4V)	40	40	µA
Synchro input voltage	90 ±10%	90 ±10%	V rms L-L
Frequency	360 min, 440 max	360 min, 440 max	Hz
Input reference voltage	115 ±10%	115 ±10%	V rms L-L
Frequency	360 min, 440 max	360 min, 440 max	Hz

6.5 Module function. The JAP module contains a complete type II servoloop, tracking, synchro to digital converter

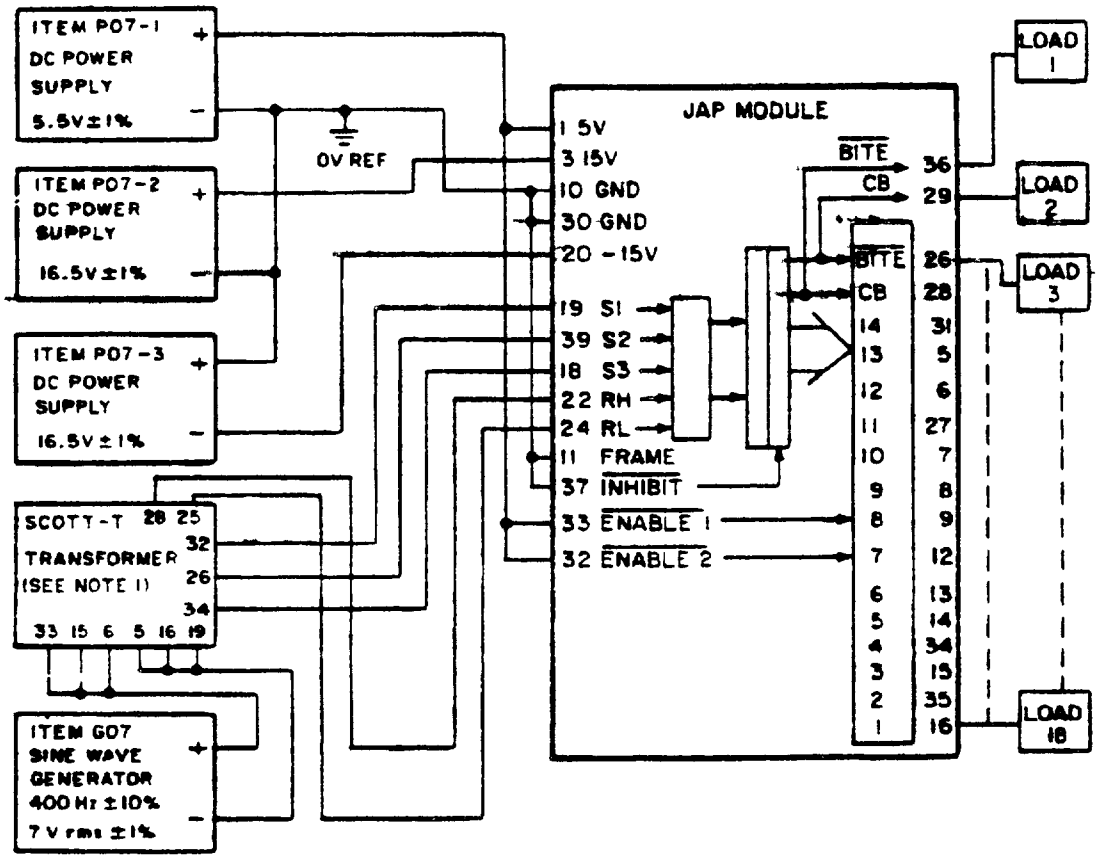
6.5.1 Input to output relationship. The JAP module accepts a 3 line, 90V line to line synchro signal and converts the angle data contained into a 14 BIT three-state digital signal. The reference is a 115V line to line signal. The module outputs a converter busy (CB) and a BITE signal which are available either directly or through the three-state buffer. The three-state buffer is controlled by 2 enable lines. ENABLE 2 controls BITS 7 through 14 and ENABLE 1 controls BITS 1 through 6, converter busy and BITE. An INHIBIT line freezes the data in the transparent latch for easy data transfer. The converter is not inhibited and continues to track the input.

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NOTES
1 Scott-T transformer = M28787/185-1 (Key Code STT)
2 Loads 1 through 18 are as shown on figure 4

FIGURE 2 Operating temperature, component temperature and life test circuit

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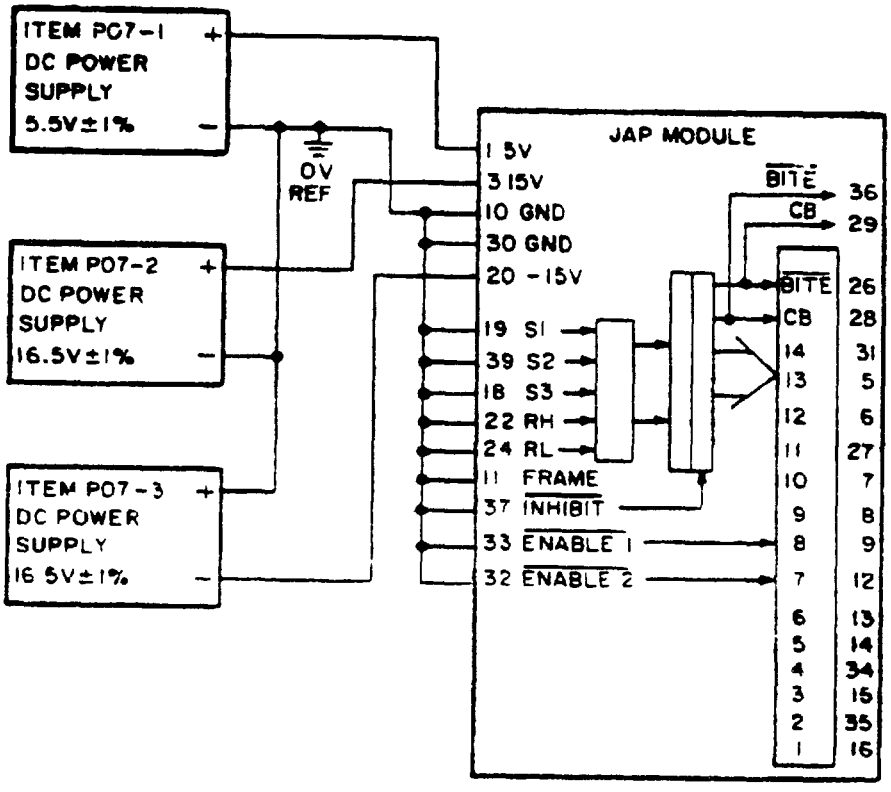
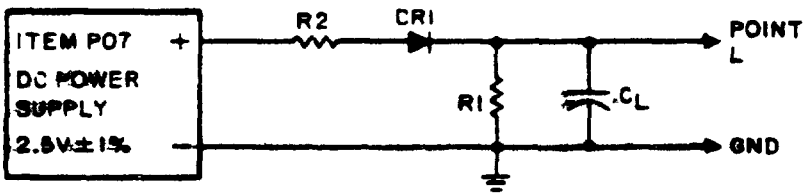


FIGURE 3 Vibration test circuit

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NOTES
1. CR1 = 1N4150 or equivalent.
2. C_L = 150 pF ±20% (including fixture and probe capacitance).
3. R1 = 60 kΩ ±1%, 1/10 watt for loads 1 and 2.
   = 6.25 kΩ ±1%, 1/10 watt for loads 3-18.
4. For loads 1 and 2, select R2 for a current flow of 1.2 mA ±5% out of point L with point L held at 0.4 V
5. For loads 3-18, select R2 for a current flow of 12 mA ±5% out of point L with point L held at 0.4 V.

FIGURE 4 JAF logic load circuit

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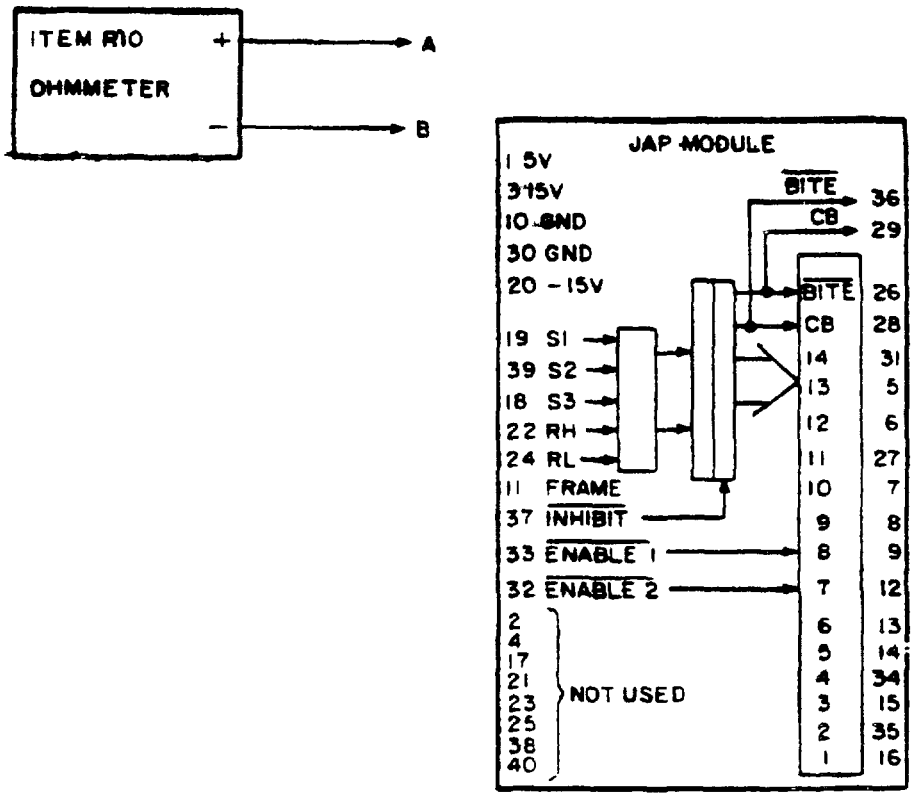
TABLE II Continuity test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A1	Continuity, frame	11	Frame	0	1	Ω	5
A2	Continuity	10	30	0	1	"	5

TABLE III. Isolation and insulation resistance test

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B1	Isolation, used to unused	1,3,5,6,7,8,9, 10 11 12,13,14, 15,16,18,19,20, 22,24,26,27 28, 29,30,31,32,33, 34,35,36,37,39	2,4,17,21, 23,25,38,40	10	∞	MΩ	6
B2	Isolation, unused to unused	2	4,17,21,23, 25,38,40	"	"	"	"
B3	"	4	17,21,23,25, 38,40	"	"	"	"
B4	"	17	21,23,25,38, 40	"	"	"	"
B5	"	21	23,25,38,40	"	"	"	"
B6	"	23	25,38,40	"	"	"	"
B7	"	25	38,40	"	"	"	"
B8	"	38	40	"	"	"	"
B9	Isolation, frame	38	11	"	"	"	"
C	Insulation resistance	18,19 22,24,39	All other pins	"	"	"	"

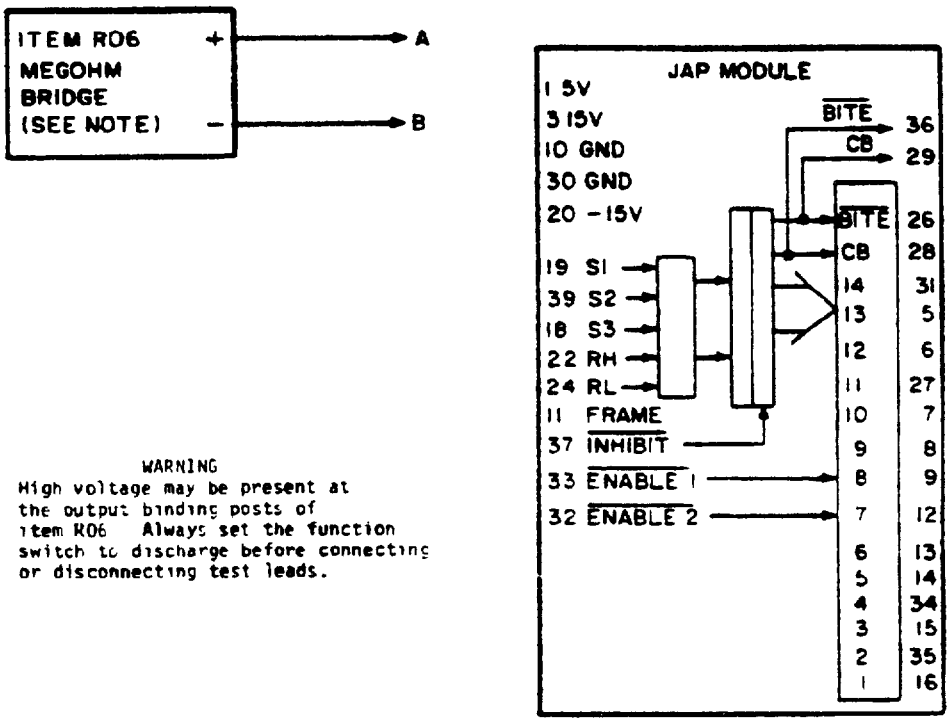
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NOTE For test condition A1, penetrate the anodized finish of the module frame with a sharply pointed probe connected to the negative terminal

FIGURE 5 Continuity test circuit for table II

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NOTE Item R06 setup
100 ±10 V dc for test conditions R1-B9.
300 ±30 V dc for test condition C.

FIGURE 6 Isolation and insulation resistance test circuit for table III

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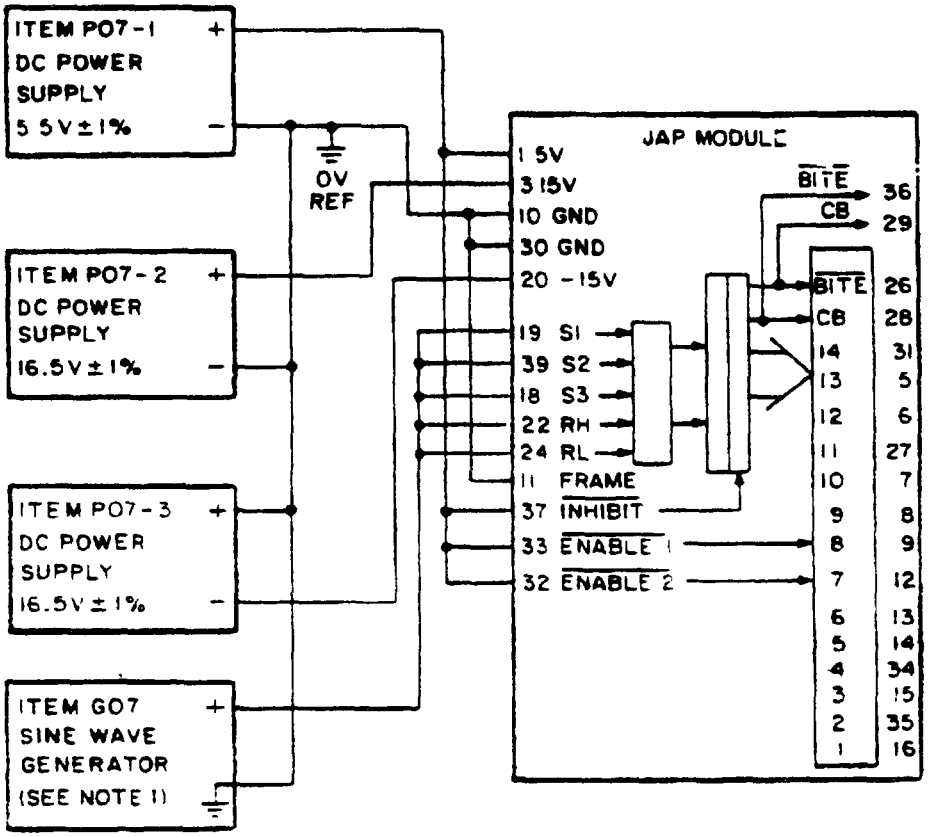
TABLE IV Common mode voltage test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
D	Common mode voltage	Survival		---	7

TABLE V. Power supply sequencing test.

Test condition	Parameter	Switch closing order			Initial limits 25°C ±5°C ambient		Unit	Test figure
		1	2	3	Min	Max		
E1	Power supply sequencing	S4	S5	S6	Survival		---	8
E2	"	S5	S6	S4	(see note 8 on figure 8)		"	"
E3	"	S6	S4	S5	"		"	"
E4	"	S6	S5	S4	"		"	"
E5	"	S5	S4	S6	"		"	"
E6	"	S4	S6	S5	"		"	"

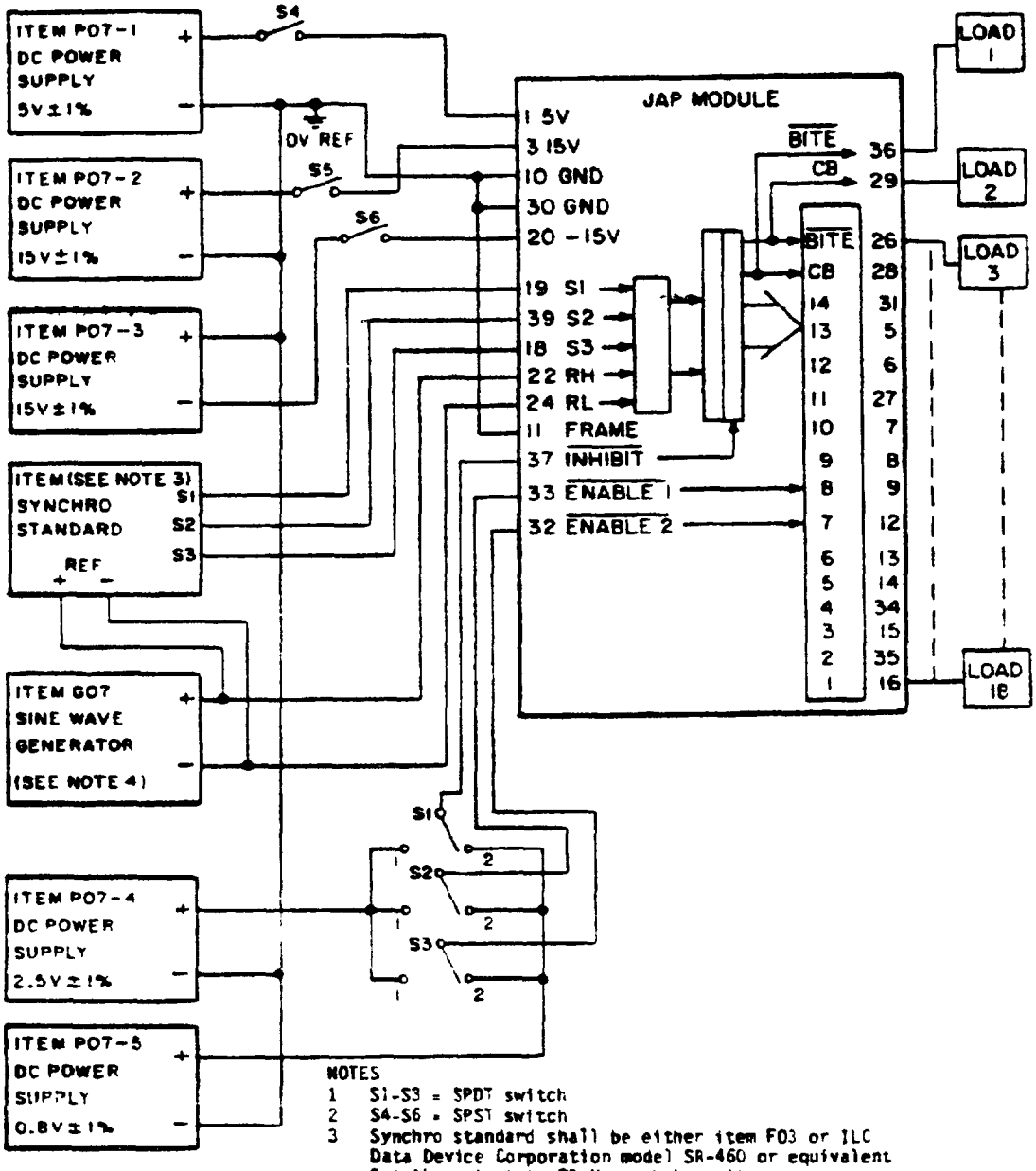
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NOTES
1 Item GO7 setup
106 V rms ±1% at 400 Hz ±10%
2 The module shall pass this test if it passes the remaining tests in the sequence of tests

FIGURE 7 Common mode voltage test circuit for table IV

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NOTES

1. S1-S3 = SPDT switch
2. S4-S6 = SPST switch
3. Synchro standard shall be either item F03 or ILC Data Device Corporation model SR-460 or equivalent Set the output to 90 V rms L-L position
4. Item GO7 setup
   126.5 V rms ±1% at 400 Hz ±10%
5. Set switches S1-S3 to position 1.
6. Switches S4-S6 shall be closed in the order specified in table V.
7. Loads 1 through 18 are as shown on figure 4
8. The module shall pass if it passes the remaining tests.

FIGURE 6 Power supply sequencing test circuit for table V

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TABLE VI Power supply current test

Test condition		Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
			Min	Max		
F1	15V	Power supply current	0.1	30 0	mA	9
F2	-15V	"	0 01	30 00	"	"
F3	5V	"	0 1	130 0	"	"

TABLE VII Power supply capacitance test

Test condition		Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
G1	5V	Power supply capacitance	1	10	1 82	3 38	µF	10
G2	15V	Power supply capacitance	3	10	1 96	3 64	µF	10

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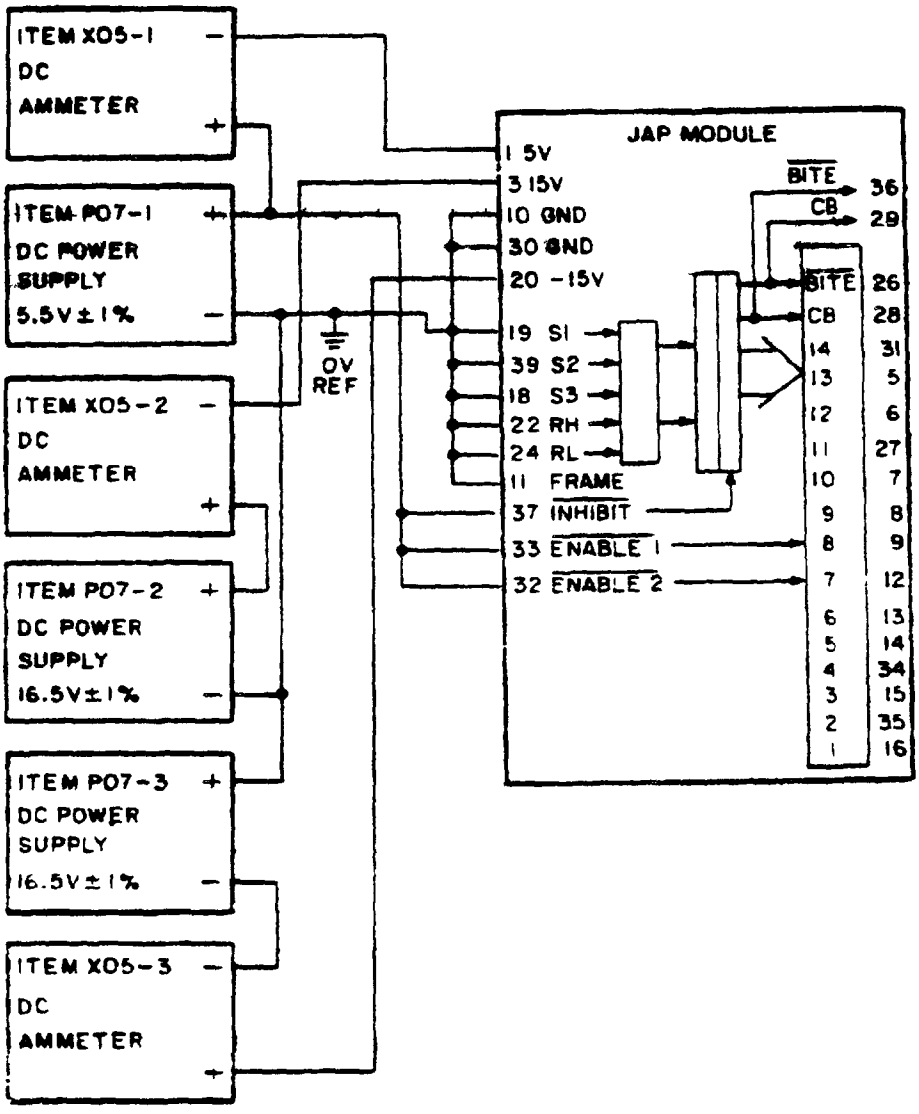
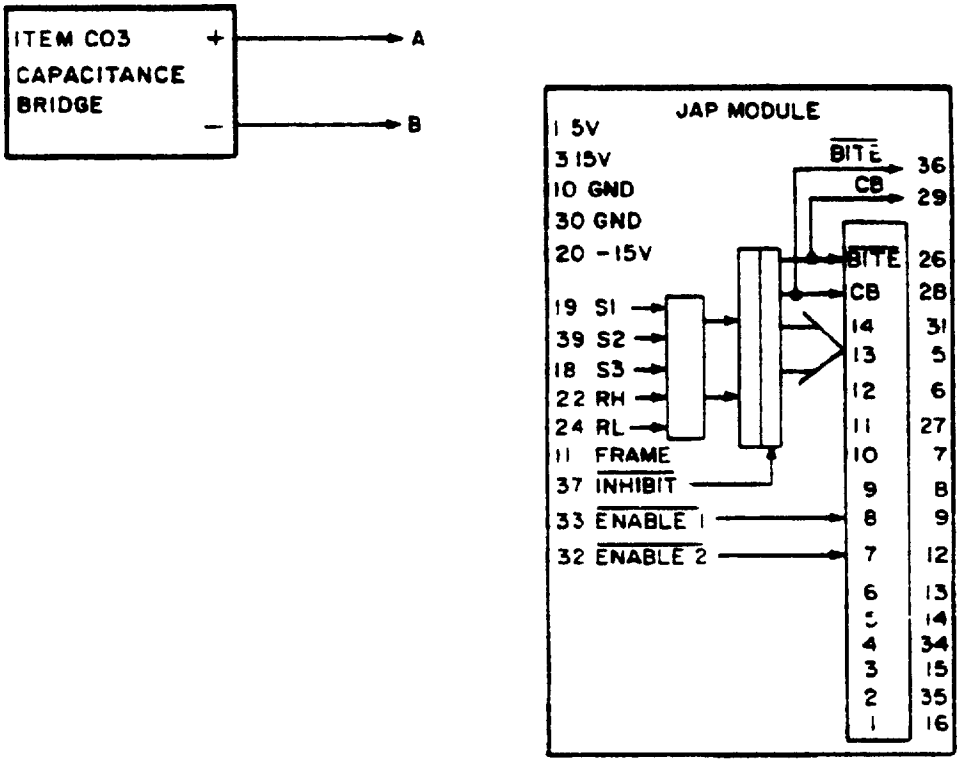


FIGURE 9 Power supply current test circuit for table VI

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NOTE Item CO3 setup
Frequency = 1 kHz ±10%
V_{pk} = 0.3 V maximum

FIGURE 10 Power supply capacitance test circuit for table VII

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TABLE VIII Static angular accuracy test

Test condition	Parameter	Angle input	Output pin states														Initial limits 25°C ±5°C ambient		Unit	Test figure
			16	35	15	34	14	13	12	9	8	7	27	6	5	31	Min	Max		
H1	Static angular accuracy, V_{normal}	000 00	L	L	L	L	L	L	L	L	L	L	L	L	L	L	-4	4	LSB	11
H2	"	015 00	L	L	L	L	H	L	H	L	H	L	H	L	H	H	"	"	"	"
H3	"	030.00	L	L	L	H	L	H	L	H	L	H	L	H	L	H	"	"	"	"
H4	"	045 00	L	L	H	L	L	L	L	L	L	L	L	L	L	L	"	"	"	"
H5	"	060.00	L	L	H	L	H	L	H	L	H	L	H	L	H	H	"	"	"	"
H6	"	075 00	L	L	H	H	L	H	L	H	L	H	L	H	L	H	"	"	"	"
H7	"	090 00	L	H	L	L	L	L	L	L	L	L	L	L	L	L	"	"	"	"
H8	"	105 00	L	H	L	L	H	L	H	L	H	L	H	L	H	H	"	"	"	"
H9	"	120 00	L	H	L	H	L	H	L	H	L	H	L	H	L	H	"	"	"	"
H10	"	135 00	L	H	H	L	L	L	L	L	L	L	L	L	L	L	"	"	"	"
H11	"	150 00	L	H	H	L	H	L	H	L	H	L	H	L	H	H	"	"	"	"
H12	"	165 00	L	H	H	H	L	H	L	H	L	H	L	H	L	H	"	"	"	"
H13	"	180.00	H	L	L	L	L	L	L	L	L	L	L	L	L	L	"	"	"	"
H14	"	225.00	H	L	H	L	L	L	L	L	L	L	L	L	L	L	"	"	"	"
H15	"	270 00	H	H	L	L	L	L	L	L	L	L	L	L	L	L	"	"	"	"
H16	"	315 00	H	H	H	L	L	L	L	L	L	L	L	L	L	L	"	"	"	"
H17	"	359.98	H	H	H	H	H	H	H	H	H	H	H	H	H	H	"	"	"	"

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TABLE IX Static angular accuracy test

Test condition	Parameter	Angle input	Output pin states														Initial limits 25°C ±5°C ambient		Unit	Test figure
			16	35	15	34	14	13	12	9	8	7	27	6	5	31	Min	Max		
J1	Static angular accuracy, V_{low}	000 00	L	L	L	L	L	L	L	L	L	L	L	L	L	L	-4	4	LSB	11
J2	"	015 00	L	H	L	L	H	L	H	L	H	L	H	L	H	H	"	"	"	"
J3	"	030 00	L	H	L	H	L	H	L	H	L	H	L	H	L	H	"	"	"	"
J4	"	045 00	L	L	H	L	L	L	L	L	L	L	L	L	L	L	"	"	"	"
J5	"	060 00	L	L	H	L	H	L	H	L	H	L	H	L	H	H	"	"	"	"
J6	"	075 00	L	L	H	H	L	H	L	H	L	H	L	H	L	H	"	"	"	"
J7	"	090 00	L	H	L	L	L	L	L	L	L	L	L	L	L	L	"	"	"	"
J8	"	105 00	L	H	L	L	H	L	H	L	H	L	H	L	H	H	"	"	"	"
J9	"	120 00	L	H	L	H	L	H	L	H	L	H	L	H	L	H	"	"	"	"
J10	"	135 00	L	H	H	L	L	L	L	L	L	L	L	L	L	L	"	"	"	"
J11	"	150 00	L	H	H	L	H	L	H	L	H	L	H	L	H	H	"	"	"	"
J12	"	165 00	L	H	H	H	L	H	L	H	L	H	L	H	L	H	"	"	"	"
J13	"	180 00	H	L	L	L	L	L	L	L	L	L	L	L	L	L	"	"	"	"
J14	"	225 00	H	L	H	L	L	L	L	L	L	L	L	L	L	L	"	"	"	"
J15	"	270 00	H	H	L	L	L	L	L	L	L	L	L	L	L	L	"	"	"	"
J16	"	315 00	H	H	H	L	L	L	L	L	L	L	L	L	L	L	"	"	"	"
J17	"	359 98	H	H	H	H	H	H	H	H	H	H	H	H	H	H	"	"	"	"

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TABLE X. Static angular accuracy test.

Test condition	Parameter	Angle input	Output pin states														Initial limits 25°C ±5°C ambient		Unit	Test figure
			16	35	15	34	14	13	12	9	8	7	27	6	5	31	Min	Max		
K1	Static angular accuracy, V_{high}	000 00	H	L	L	L	L	L	L	L	L	L	L	L	L	L	-4	4	LSB	11
K2	"	015 00	L	L	L	L	H	L	H	L	H	L	H	L	H	H	"	"	"	"
K3	"	030 00	L	L	L	H	L	H	L	H	L	H	L	H	L	H	"	"	"	"
K4	"	045 00	L	L	H	L	L	L	L	L	L	L	L	L	L	L	"	"	"	"
K5	"	060 00	L	L	H	L	H	L	H	L	H	L	H	L	H	H	"	"	"	"
K6	"	075 00	L	L	H	H	L	H	L	H	L	H	L	H	L	H	"	"	"	"
K7	"	090 00	L	H	L	L	L	L	L	L	L	L	L	L	L	L	"	"	"	"
K8	"	105 00	L	H	L	L	H	L	H	L	H	L	H	L	H	H	"	"	"	"
K9	"	120 00	L	H	L	H	L	H	L	H	L	H	L	H	L	H	"	"	"	"
K10	"	135 00	L	H	H	L	L	L	L	L	L	L	L	L	L	L	"	"	"	"
K11	"	150 00	L	H	H	L	H	L	H	L	H	L	H	L	H	H	"	"	"	"
K12	"	165 00	L	H	H	H	L	H	L	H	L	H	L	H	L	H	"	"	"	"
K13	"	180 00	H	L	L	L	L	L	L	L	L	L	L	L	L	L	"	"	"	"
K14	"	225 00	H	L	H	L	L	L	L	L	L	L	L	L	L	L	"	"	"	"
K15	"	270 00	H	H	L	L	L	L	L	L	L	L	L	L	L	L	"	"	"	"
K16	"	315 00	H	H	H	L	L	L	L	L	L	L	L	L	L	L	"	"	"	"
K17	"	359 98	H	H	H	H	H	H	H	H	H	H	H	H	H	H	"	"	"	"

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TABLE XI Sensitivity check test

Test condition	Parameter	Angle input	Output pin states																Initial limits 25°C ±5°C ambient		Unit	Test figure
			16	15	15	34	14	13	12	9	8	7	27	6	5	31			Min	Max		
L	Sensitivity check	45 00	I	L	H	L	L	L	L	L	L	L	L	L	L	L			0.010	0.033	Degree	11

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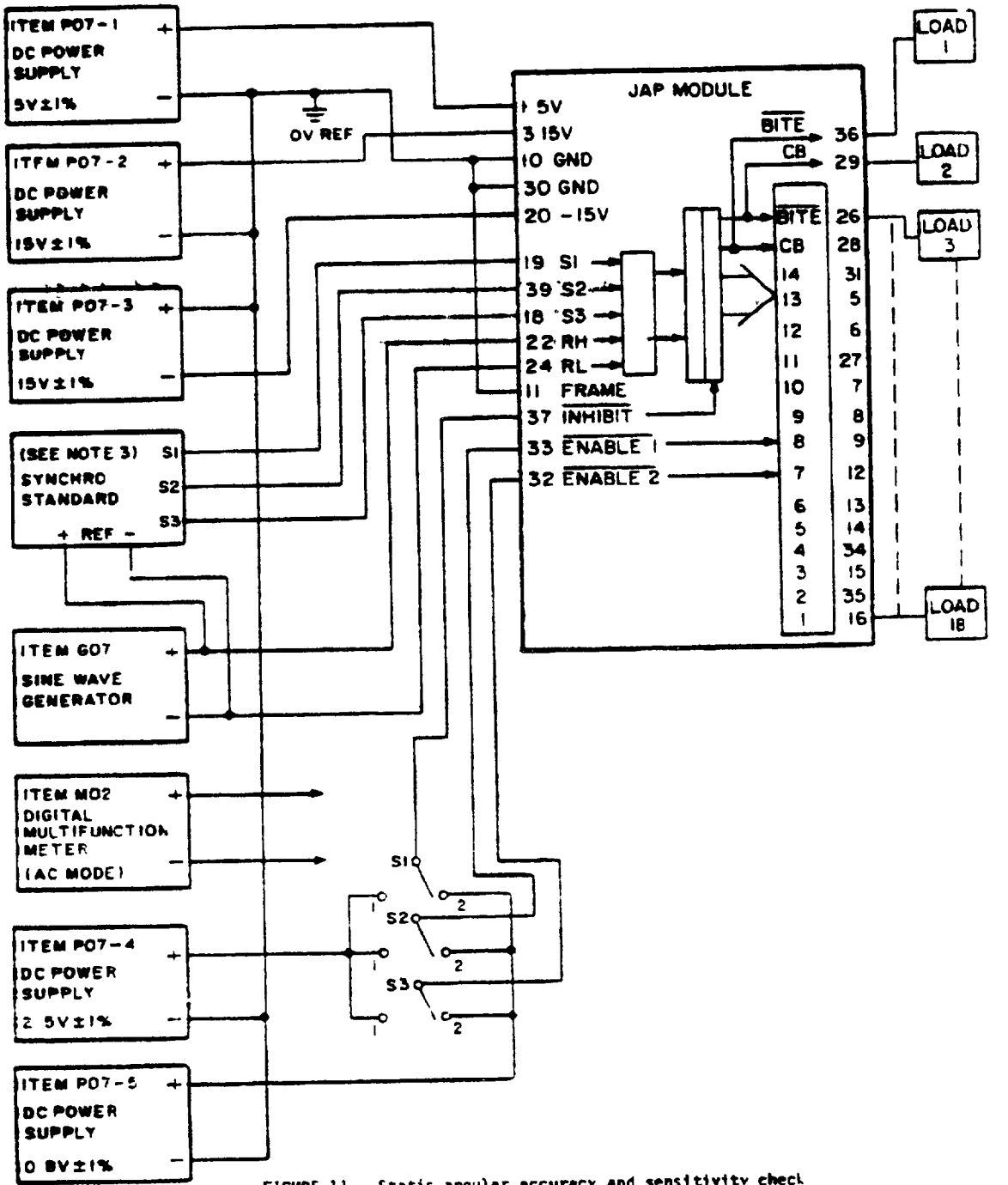


FIGURE 11 Static angular accuracy and sensitivity check test circuit for tables VIII through XI

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NOTES:
1. S1 = S2 = S3 = SPDT switch.
2. S1 shall be in position 1 and S2 and S3 in position 2.
3. Synchro standard is ILC Data Device Corporation model SR-460 or equivalent.
4. The synchro standard controls shall be set to output a 90 V rms L-L synchro signal.
5. For test conditions H1-H17, set synchro standard for 400 Hz and adjust item G07 for 400 Hz ±10%, 115 V rms ±1%. Use item M02 to adjust voltage.
6. Set the synchro standard to the angle specified in the test table and monitor the module outputs. These outputs shall be as specified in the test table
7. For test conditions J1-J17, adjust item G07 to 103.5 V rms ±1% and repeat note 6.
8. For test conditions K1-K17, adjust item G07 to 126.5 V rms ±1% and repeat note 6
9. For test condition L, set synchro standard for 400 Hz and adjust item G07 for 115 V rms ±1% at 400 Hz ±1%. Adjust synchro standard to obtain the module output angle specified in table XI. Increase the synchro standard setting until the module digital output increases +1 LSB. Record the synchro standard setting. Again increase the synchro standard setting until the module digital output increases an additional +1 LSB. Record the synchro standard setting. The difference of the two settings shall meet the limits of table XI, test condition L.
10. Loads 1 through 18 are as shown on figure 4

FIGURE 11. Static angular accuracy and sensitivity check test circuit for tables VIII through XI - Continued

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TABLE XII. Digital output voltage (V_{OH}) test.

Test condition	Parameter		Input angle	Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
M1	BIT 1	Digital output voltage (V_{OH})	359 98°	16	2.4	5.1	V	12
M2	BIT 2	"	"	35	"	"	"	"
M3	BIT 3	"	"	15	"	"	"	"
M4	BIT 4	"	"	34	"	"	"	"
M5	BIT 5	"	"	14	"	"	"	"
M6	BIT 6	"	"	13	"	"	"	"
M7	BIT 7	"	"	12	"	"	"	"
M8	BIT 8	"	"	9	"	"	"	"
M9	BIT 9	"	"	8	"	"	"	"
M10	BIT 10	"	"	7	"	"	"	"
M11	BIT 11	"	"	27	"	"	"	"
M12	BIT 12	"	"	6	"	"	"	"
M13	BIT 13	"	"	5	"	"	"	"
M14	BIT 14	"	"	31	"	"	"	"

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TABLE XIII. Digital output voltage (V_{OL}) test.

Test condition	Parameter	Input angle	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	Min	Max		
N1	BIT 1 Digital output voltage (V_{OL})	000 00°	16	0	400	mV	12
N2	BIT 2 "	"	35	"	"	"	"
N3	BIT 3 "	"	15	"	"	"	"
N4	BIT 4 "	"	34	"	"	"	"
N5	BIT 5 "	"	14	"	"	"	"
N6	BIT 6 "	"	13	"	"	"	"
N7	BIT 7 "	"	12	"	"	"	"
N8	BIT 8 "	"	9	"	"	"	"
N9	BIT 9 "	"	8	"	"	"	"
N10	BIT 10 "	"	7	"	"	"	"
N11	BIT 11 "	"	27	"	"	"	"
N12	BIT 12 "	"	6	"	"	"	"
N13	BIT 13 "	"	5	"	"	"	"
N14	BIT 14 "	"	31	"	"	"	"

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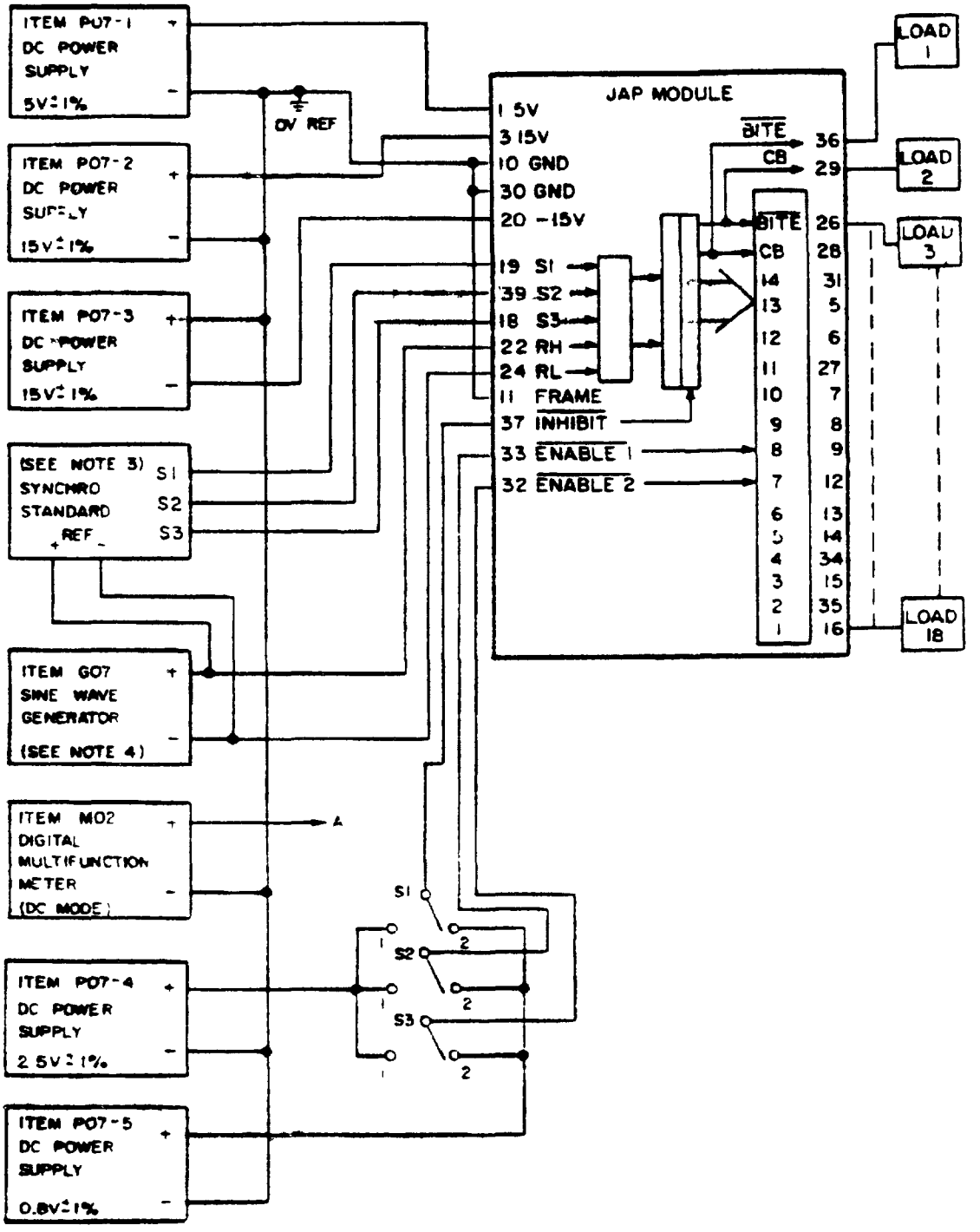


FIGURE 1. Digital output voltage (V_{OH} and V_{OL}) test circuit for tables XII and XIII.

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NOTES
1. S1 = S2 = S3 = SPDT switch
2. S1 shall be in position 1 and S2 and S3 shall be in position 2.
3. Synchro standard is ILC Data Device Corporation model SR-460 or equivalent. The output shall be set to 90 V rms L-L position.
4. Item G07 setup.
   115 V rms ±10% at 400 Hz ±10%.
5. It may be necessary to slightly adjust the synchro standard setting to obtain the desired output state.
6. Loads 1 through 18 are as shown on figure 4.

FIGURE 12. Digital output voltage (V_{OH} and V_{OL} test circuit for tables XII and XIII - Continued

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TABLE XIV Digital input breakdown current test

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
P1	ENABLE 1 Digital input breakdown current	Open	33	32,37	0	40	uA	13
P2	ENABLE 2 "	Open	32	33,37	0	40	uA	13
P3	INHIBIT "	Open	37	32,33	0	40	uA	13

TABLE XV. Digital input current test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
Q1	ENABLE 1 Digital input current	Open	33	32,37	-1 6	0.0	mA	13
Q2	ENABLE 2 "	Open	32	33,37	-1 6	0 0	mA	13
Q3	INHIBIT "	Open	37	32,33	-1 6	0 0	mA	13

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TABLE XVI Three-state leakage current (high level) test

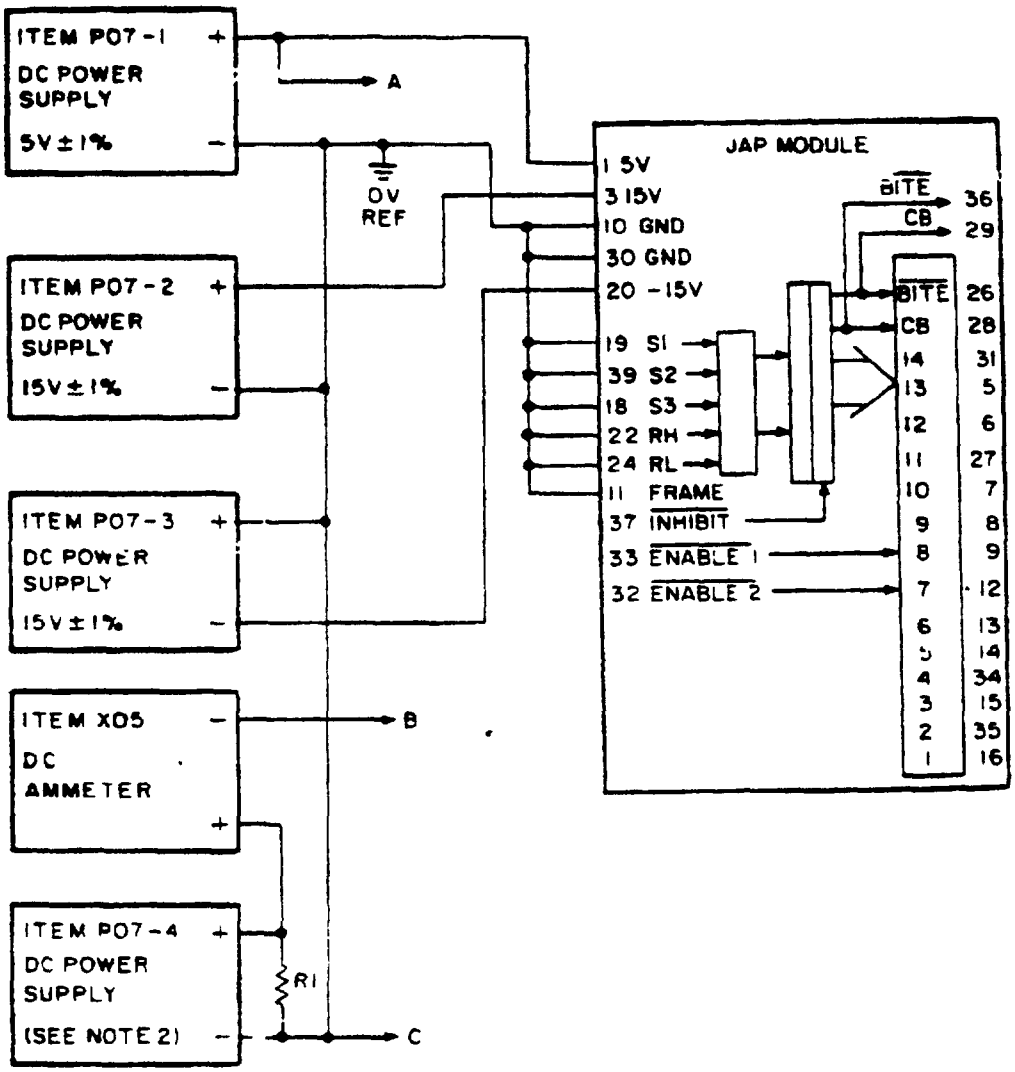
Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
R1	BIT1 Three-state leakage current, high level	32,33	16	37	-1	20	uA	13
R2	BIT2 "	"	35	"	"	"	"	"
R3	BIT3 "	"	15	"	"	"	"	"
R4	BIT4 "	"	34	"	"	"	"	"
R5	BIT5 "	"	14	"	"	"	"	"
R6	BIT6 "	"	13	"	"	"	"	"
R7	BIT7 "		12	"	"	"	"	"
R8	BIT8 "	"	9	"	"	"	"	"
R9	BIT9 "	"	8		"	"	"	"
R10	BIT10 "	"	7	"	"	"	"	"
R11	BIT11 "	"	27	"	"	"	"	"
R12	BIT12 "	"	6	"	"	"	"	"
R13	BIT13 "	"	5	"	"	"	"	"
R14	BIT14 "	"	31	"	"	"	"	"
R15	CB "	"	28		"	"	"	"
R16	BITE "	"	26	"	"	"	"	"

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TABLE XVII. Three-state leakage current (low-level) test

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
S1	BIT1 Three-state leakage current, low level	32,33	16	37	1	-20	µA	13
S2	BIT2 "	"	35	"	"	"	"	"
S3	BIT3 "	"	15	"	"	"	"	"
S4	BIT4 "	"	34	"	"	"	"	"
S5	BIT5 "	"	14	"	"	"	"	"
S6	BIT6 "	"	13	"	"	"	"	"
S7	BIT7 "	"	12	"	"	"	"	"
S8	BIT8 "	"	9	"	"	"	"	"
S9	BIT9 "	"	8	"	"	"	"	"
S10	BIT10 "	"	7	"	"	"	"	"
S11	BIT11 "	"	27	"	"	"	"	"
S12	BIT12 "	"	6	"	"	"	"	"
S13	BIT13 "	"	5	"	"	"	"	"
S14	BIT14 "	"	31	"	"	"	"	"
S15	CB "	"	28	"	"	"	"	"
S16	$\overline{\text{BITE}}$ "	"	26	"	"	"	"	"

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NOTES
1. R1 = 420Ω ±1%, 1/8 W
2. Adjust item P07-4 voltages as follows:
   5.5 V ±1% for test conditions P1-P3
   0.4 V ±1% for test conditions Q1-Q3 and S1-S16.
   2.7 V ±1% for test conditions R1-R16.

FIGURE 10. Digital input breakdown current, digital input current and three-state leakage current (high-level and low-level) test circuit for tables XIV through XVII.

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TABLE XVIII Delay time (INHIBIT, ENABLE 2 and ENABLE 1) test

Test condition	Parameter	Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
T	Delay time, INHIBIT	Open	32,33	37	37	31	0	3	us	14,15
U1	Delay time, ENABLE 2	37	Open	32	32	31	"	150	ns	14,16
U2	Delay time, ENABLE 1	37	Open	33	33	13	"	150	ns	14,16

TABLE XIX Tracking rate check test

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
V1	Tracking rate check, 400 Hz, count down	37	32,33,D	36	360	540	kHz	18
V2	Tracking rate check 400 Hz, count up	37,D	32,33	36	360	540	kHz	18

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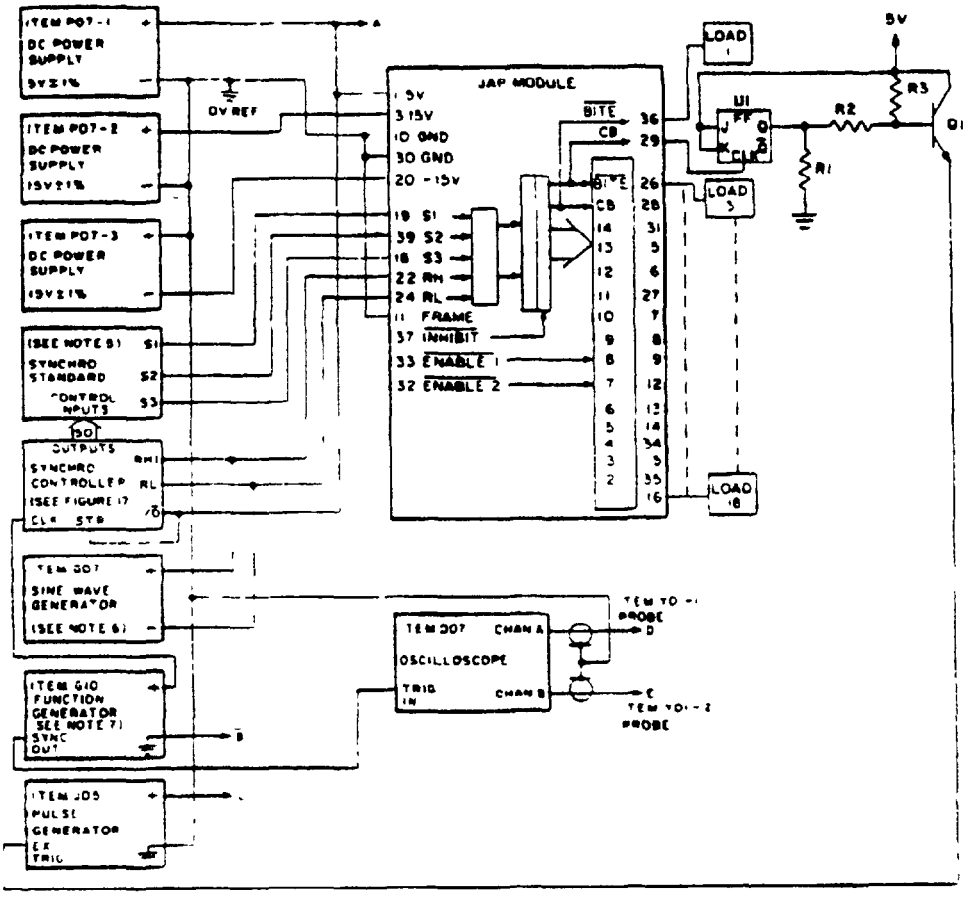


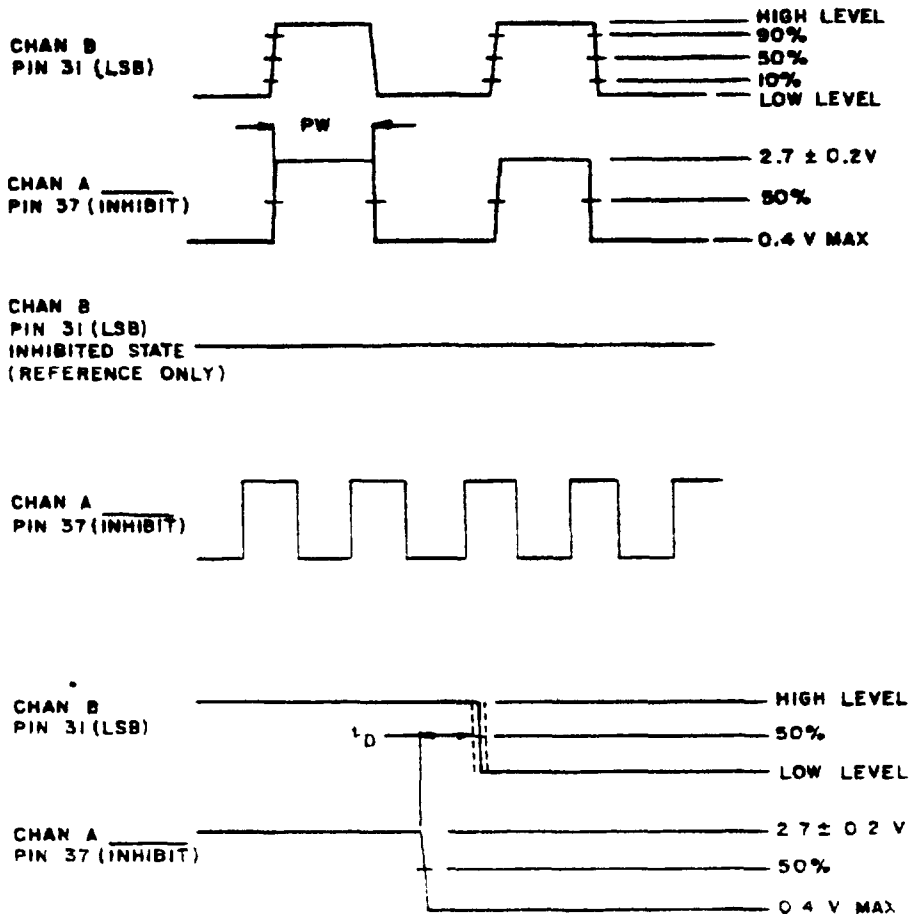
FIGURE 14 Delay time (INHIBIT, ENABLE 2, and ENABLE 1) test circuit for table XVIII.

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NOTES.
1. R1 = R3 = 1kΩ ±5%, 1/10 W.
2. R2 = 100Ω ±5%, 1/10 W.
3. Q1 = 2N2222A or equivalent.
4. U1 = SN5470 or 54LS70 or equivalent.
5. Synchro standard is ILC Data Device Corporation model SR-460 or equivalent. The output voltage shall be set to 90 V rms L-L position.
6. Set item G07 to 115 V rms ±10% at 400 Hz ±10%.
7. Procedure for test condition T.
   a. Connect the circuit as specified in table XVIII.
   b Adjust item G10 to 3.5 V ±10% at 100 kHz ±10%.
   c. Set item J05 to external trigger and adjust the pulse width and amplitude as specified on figure 15.
   d. Slowly adjust the delay of item J05 so that the negative going edge of the waveform on pin 37 lags the negative edge of the waveform on pin 31. Continue to increase the delay between the waveforms until the LSB stops switching. Slowly back off the delay until the switching point is again discernible. Measure this delay time (t_D) as shown on figure 15. The delay time shall meet the limits in table XVIII.
8. For test condition U, set the synchro standard to "LOCAL" control and the angle to 45 degrees. Set item J05 to "NORMAL" and adjust to obtain the waveform specified on figure 16. Connect the circuit as specified in table XVIII and measure the time delay (t_{FD}) as specified on figure 16. The delay time shall meet the limits in table XVIII.
9. Loads 1 and 3 thru 18 are as shown on figure 18

FIGURE 14. Delay time ($\overline{\text{INHIBIT}}$, $\overline{\text{ENABLE 2}}$, and $\overline{\text{ENABLE 1}}$) test circuit for table XVIII - Continued.

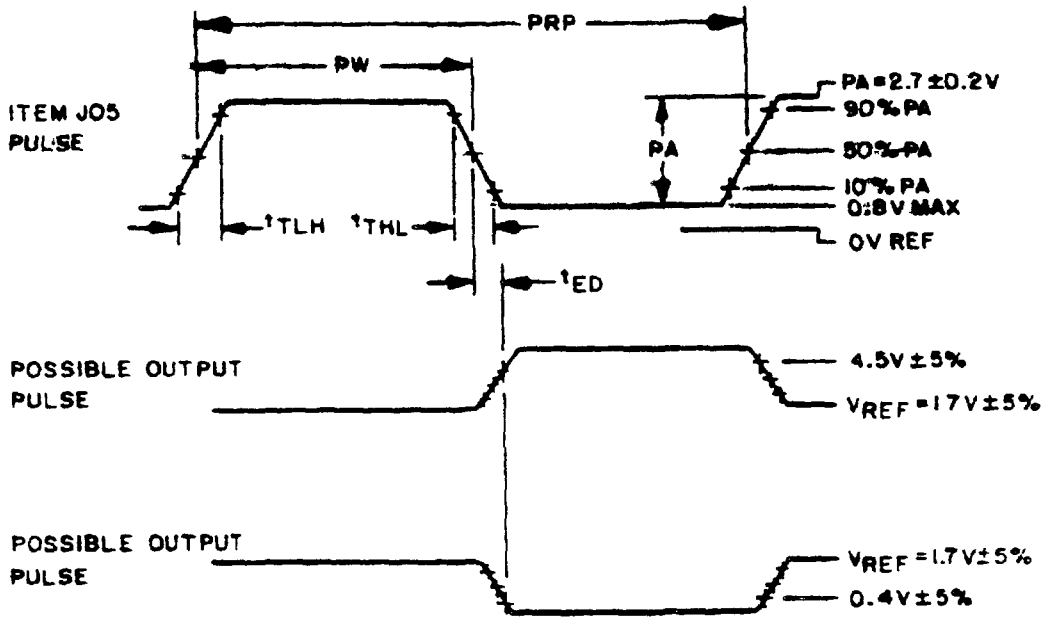
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NOTES
1. Pw = 20.0 ± 0.2 us
2. $t_{TLH} = t_{THL} < 20$ ns.
3. The center waveforms show LSB (pin 31) in inhibited state for reference only

FIGURE 15. Delay time (INHIBIT) test circuit waveforms for table XVIII.

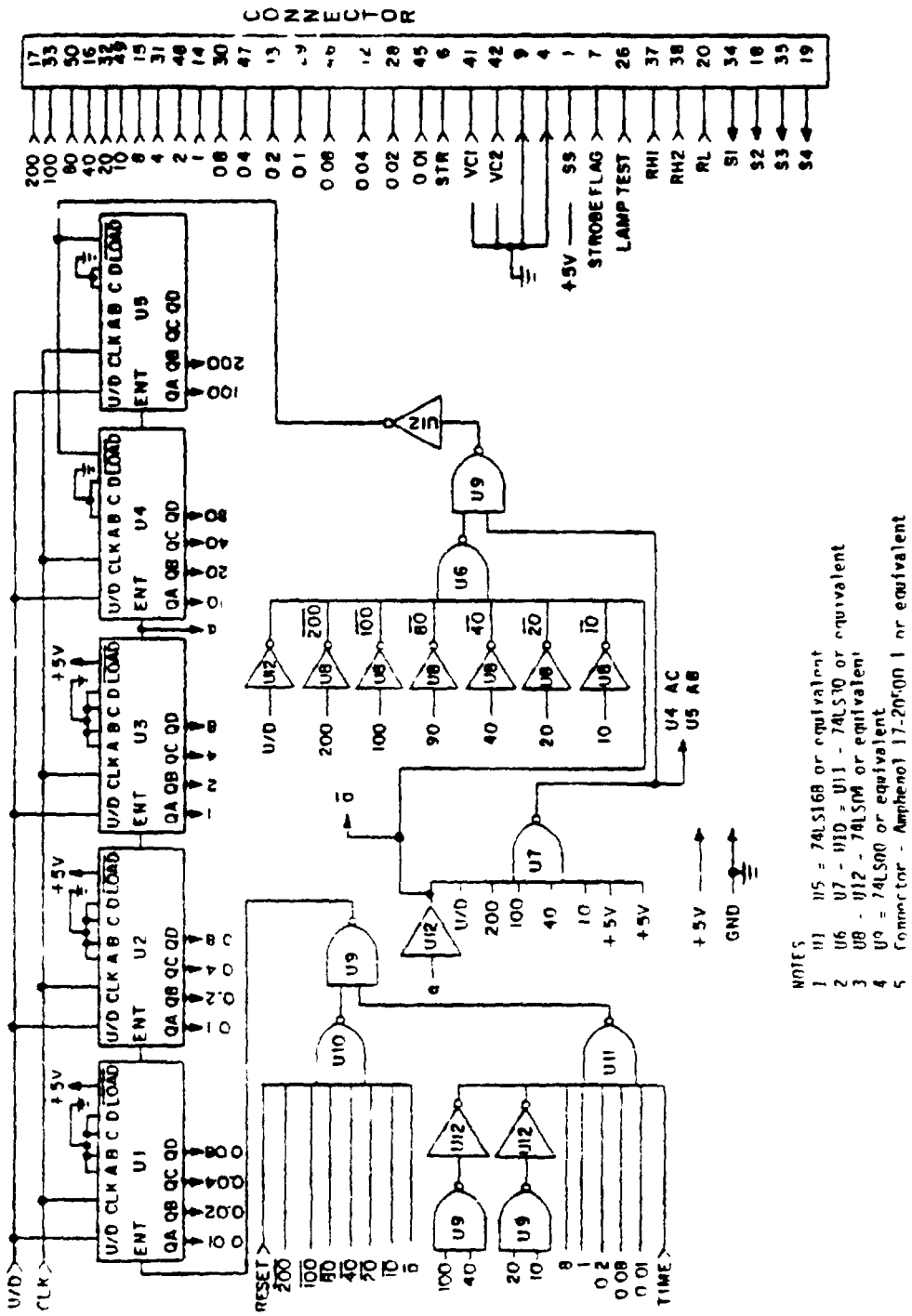
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NOTES
1. Item J05 setup
   PW = 250 ± µs
   PRP = 500 ±5 µs.
   t_{TLH} = t_{THL} < 20 ns.
2. The enable delay time (t_{ED}) shall meet the limits as specified in table XVIII.
3. Either output pulse is possible dependent upon state of the module.

FIGURE 16 Delay time ($\overline{\text{ENABLE 2}}$ and $\overline{\text{ENABLE 1}}$) test circuit waveforms for table XVIII.

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NOTES
1. U1, U5 = 74LS168 or equivalent
2. U6, U7 = U10 = U11 = 74LS30 or equivalent
3. U8 = U12 = 74LS04 or equivalent
4. U9 = 74LS00 or equivalent
5. Connector = Amphenol 17-20F00 1 or equivalent

FIGURE 17. Synchro controller

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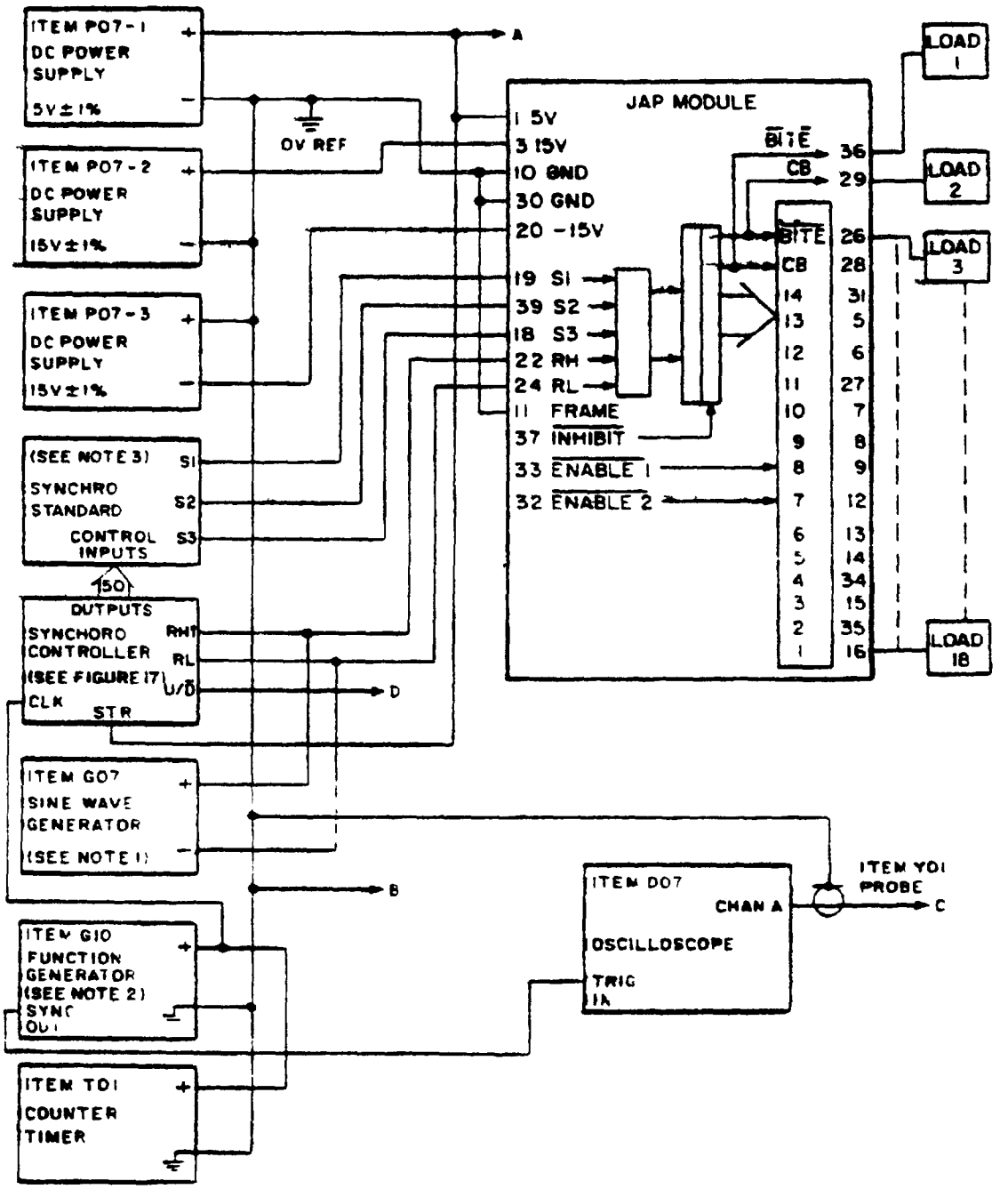


FIGURE 18. Tracking rate check test circuit for table XIX

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NOTES.
1. Item G07 setup
   115 V rms L-L ±10% at 400 Hz ±1%.
2. Adjust item G10 as follows:
   PRR = 36 kHz ±10%.
   PW = 6 µs ±10%.
   PA = 3.5 V ±10%.
   $t_{TLH} = t_{THL} \leq 100$ ns.
3. Synchro standard is ILC Data Device Corporation model SR-460 or equivalent.
   Set the output at 90 V rms L-L ±1% at 400 Hz ±10%.
4. Verify that pin 36 $\overline{\text{BITE}}$ is in the high level state (≥ 2.4 V) Increase the frequency of item G10 until pin 36 output voltage level goes to the low level state (≤ 0.4 V). This frequency as measured by item T01 shall meet the limits of table XIX.
5. The table I limits for test conditions V1 and V2 shall be calculated as follows

$$\text{TABLE I limits} = \frac{\text{TABLE XIX limits}}{36 \text{ kHz}}$$

6. This test verifies a $\overline{\text{BITE}}$ low level output voltage (V_{OL}) of 0.4 V maximum.
7. Loads 1 thru 18 are as shown on figure 4

FIGURE 18 Tracking rate check test circuit for table XIX - Continued.

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TABLE XX Settling time test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
W	Settling time	37	32,33	31	0	200	ms	19,20

TABLE XXI Noise sensitivity test

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
X	Noise sensitivity	-1	+1	LSB	21

TABLE XXII Converter busy test

Test condition	Parameter		Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	Min	Max		
Y1	Direct (V_H)	Converter busy	29	2.4	5.1	V	22,23
Y2	Direct (V_L)	"	29	0	400	mV	"
Y3	Direct (P_W)	"	29	1.5	6.0	µs	"
Y4	3-state (V_H)	"	28	2.4	5.1	V	"
Y5	3-state (V_L)	"	28	0	400	mV	"
Y6	3-state (P_W)	"	28	1.5	6.0	µs	"

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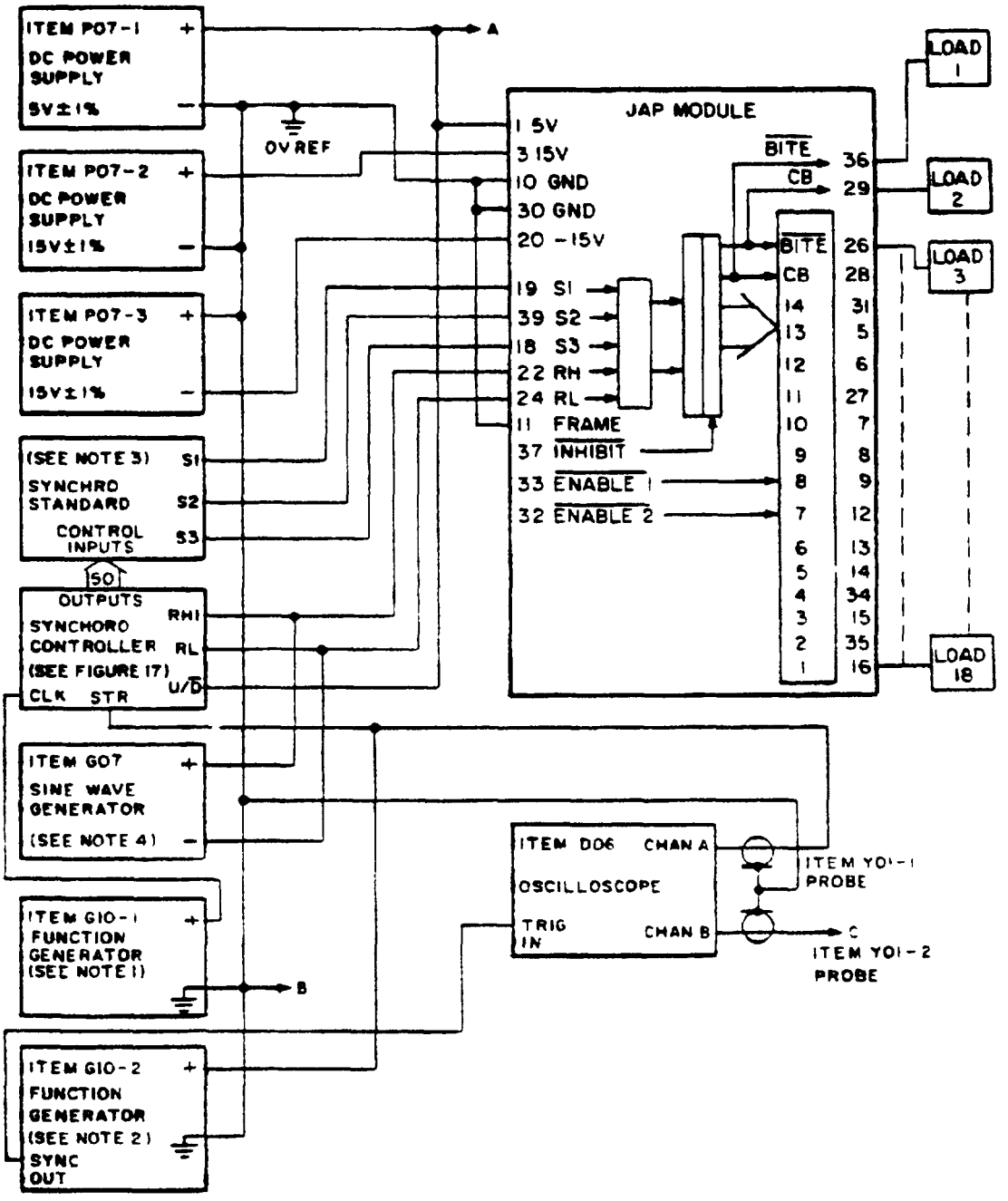


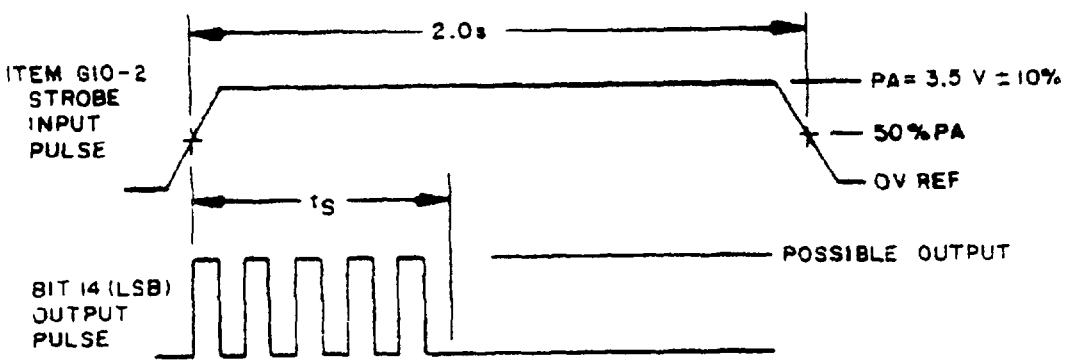
FIGURE 19 Settling time test circuit for table XX

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NOTES.
1. Item G10-1 setup
   PA = 3.5 V ±10%.
   PW = 27 µs ±10%.
   PRR = 18 kHz ±10%.
   $t_{TLH} = t_{THL} \leq 100$ ns.
2. Item G10-2 setup.
   PA = 3.5 V ±10%.
   PW = 2 µs ±10%.
   Frequency = manual trigger.
   $t_{TLH} = t_{THL} \leq 100$ ns.
3. Synchro standard is ILC Data Device Corporation model SR-460 or equivalent.
4. Item G07 setup.
   115 V rms L-L ±10% at 400 Hz ±1%.
5. Set the synchro controller reset to high and the S time to low.
6. Press the manual trigger button on item G10-2 and allow a full cycle. Verify that the angle displayed on the synchro standard is 000.00 degrees.
7. Set the synchro controller reset to low and the S time to high.
8. Press the manual trigger button on item G10-2 and measure the settling time as shown on figure 20. Verify that the angle displayed on the synchro standard is 179.29 degrees.
9. Loads 1 thru 18 are as shown on figure 4

FIGURE 19. Settling time test circuit for table XX - Continued.

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NOTE Output may settle in the high or low level.

FIGURE 20. Settling time test circuit waveforms for table XX.

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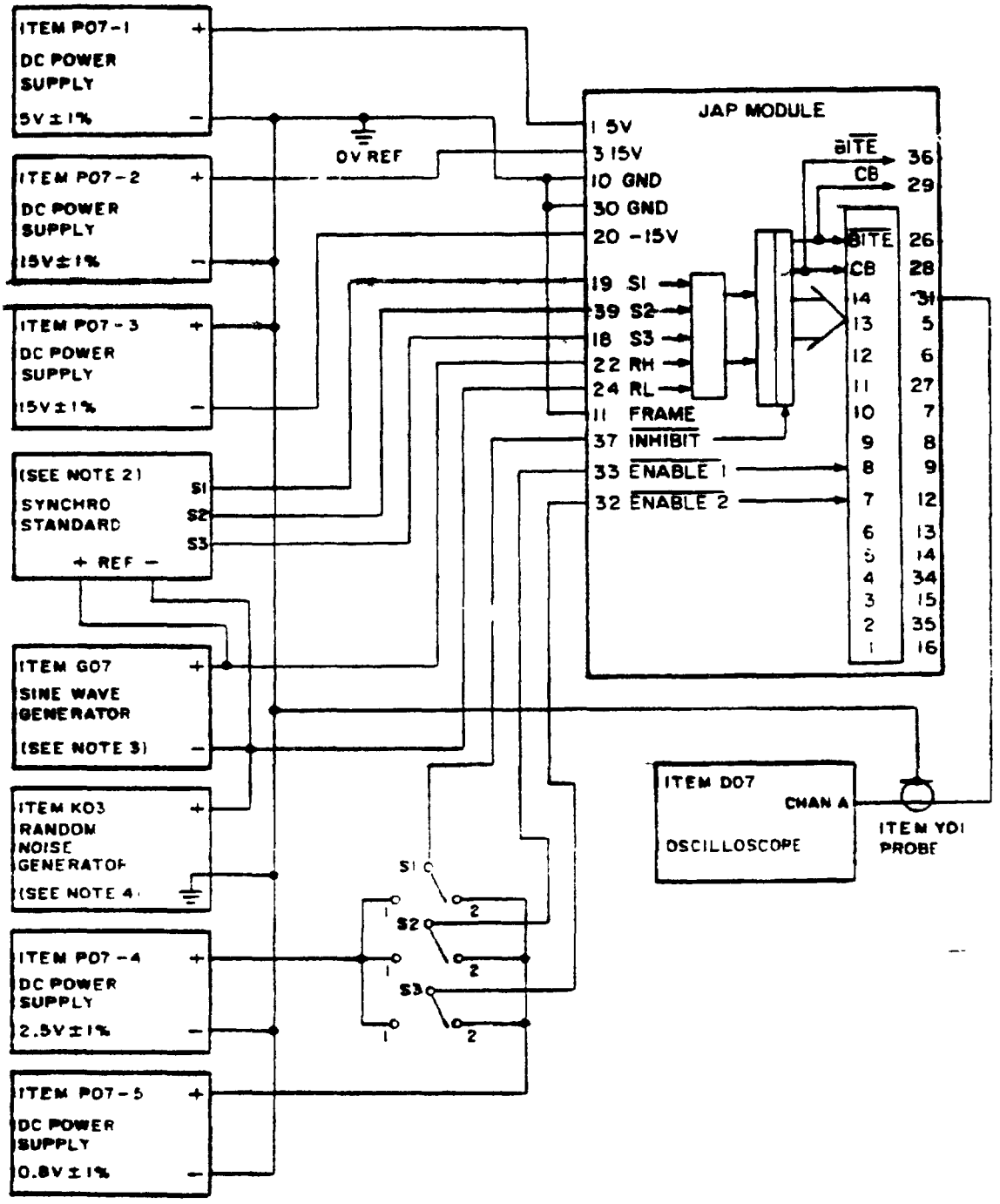


FIGURE 2. Noise sensitivity test circuit for table XX.

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NOTES:
1. S1-S3 = SPDT switch.
2. Synchro standard is ILC Data Device Corporation model SR-460 or equivalent. The output voltage shall be set to 90 V rms L-L position. The angle shall be set to 45.000 degrees.
3. Item G07 setup
   115 V rms ±10% at 400 Hz ±10%.
4. Item K03 setup:
   5 Hz to 20 kHz.
   1 V rms ±10%.
5. Set switch S1 to position 1, S2 and S3 to position 2.
6 Monitor BIT 14 (LSB). The module passes this test if the output remains stable and does not toggle.

FIGURE 21. Noise sensitivity test circuit for table XXI - Continued

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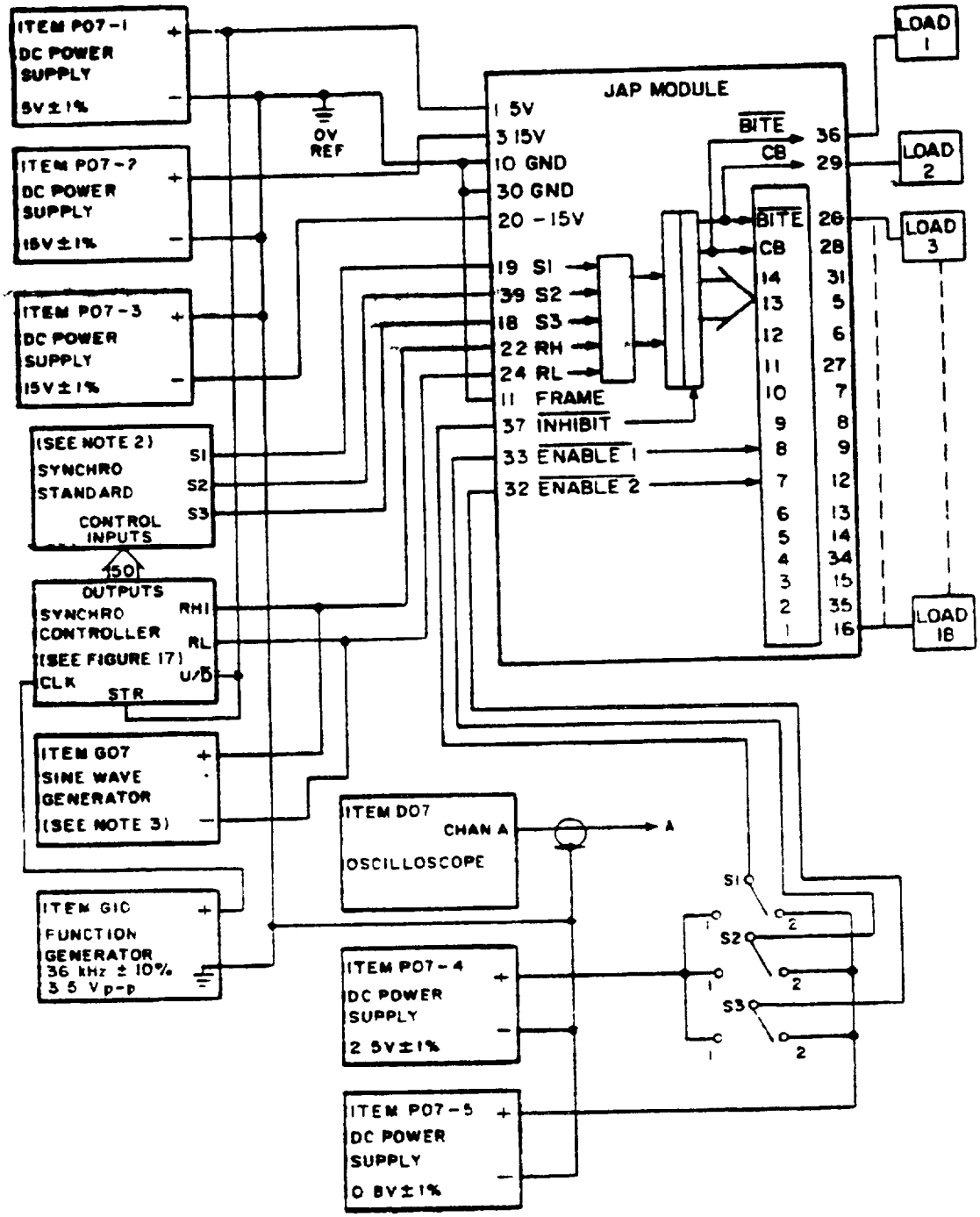


FIGURE 22 Converter busy test circuit for table XXII

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NOTES
1. S1-S3 = SPDT switch.
2. Synchro standard shall be ILC Data Device Corporation model SR-460 or equivalent. The output voltage shall be set to 90 V rms L-L position.
3. Item G07 setup.
   115 V rms ±10% at 400 Hz ±10%.
4. S1 shall be in position 1, S2 and S3 in position 2.
5. Loads 1 thru 18 are as shown on figure 4.
6. Measure the converter busy pulse parameters as shown on figure 23.

FIGURE 22. Converter busy test circuit for table XXII - Continued.

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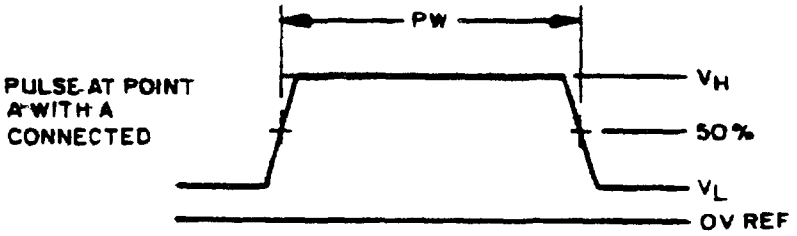


FIGURE 23. Converter busy test circuit waveform for table XXII.

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TABLE XXIII Power supply ripple rejection test

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
Z1	5V Power supply ripple rejection	1	Open	1	1	LSB	24
Z2	15V "	1	Open	"	"	"	"
Z3	15V "	Open	20	"	"	"	"

TABLE XXIV Continuous conversion test

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
AA	Continuous conversion	See figure 25 note 6		---	25

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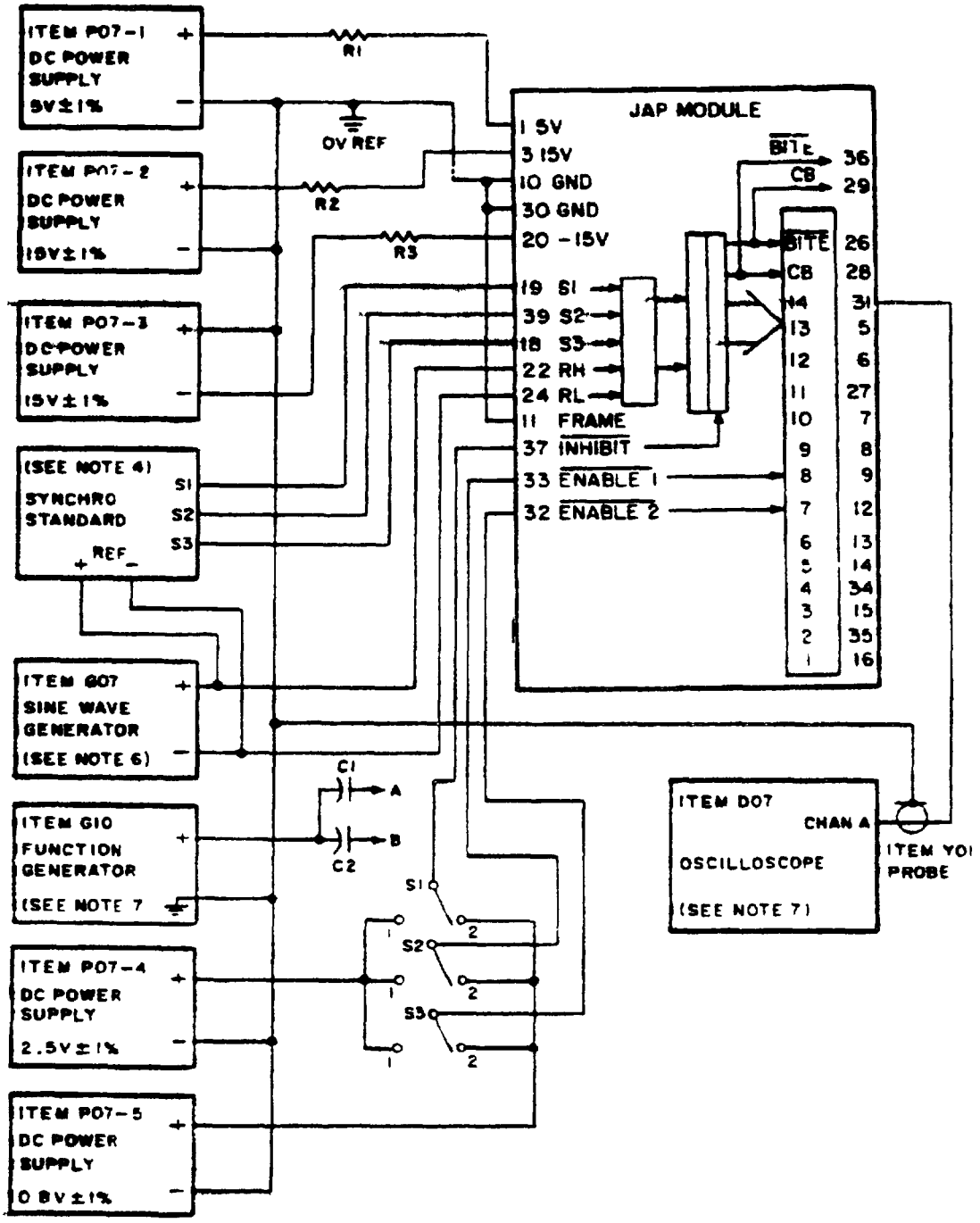


FIGURE 24. Power supply ripple rejection test circuit for table XXII.

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NOTES:
1. R1-R3 = 100Ω ±5%, 1/8 W.
2. C1 = C2 = 20 µf 20%, 20 V dc.
3. S1-S3 = SPDT switch.
4. Synchro standard is ILC Data Device Corporation model SR-460 or equivalent. The output voltage shall be set to 90 V rms L-L position. The angle shall be set to 45.00 degrees.
5. Set S1 to position 1 and S2 and S3 to position 2.
6. Item G07 setup:
   115 V rms ±10% at 400 Hz ±10%.
7. Item G10 setup, 1 kHz ±1% at:
   0.5 V rms ±1% for test condition Z1.
   1 V rms ±1% for test conditions Z2 and Z3.
8. Monitor item D07. The module shall pass the test if the output does not toggle.

FIGURE 24. Power supply ripple rejection test circuit for table XXIII - Continued

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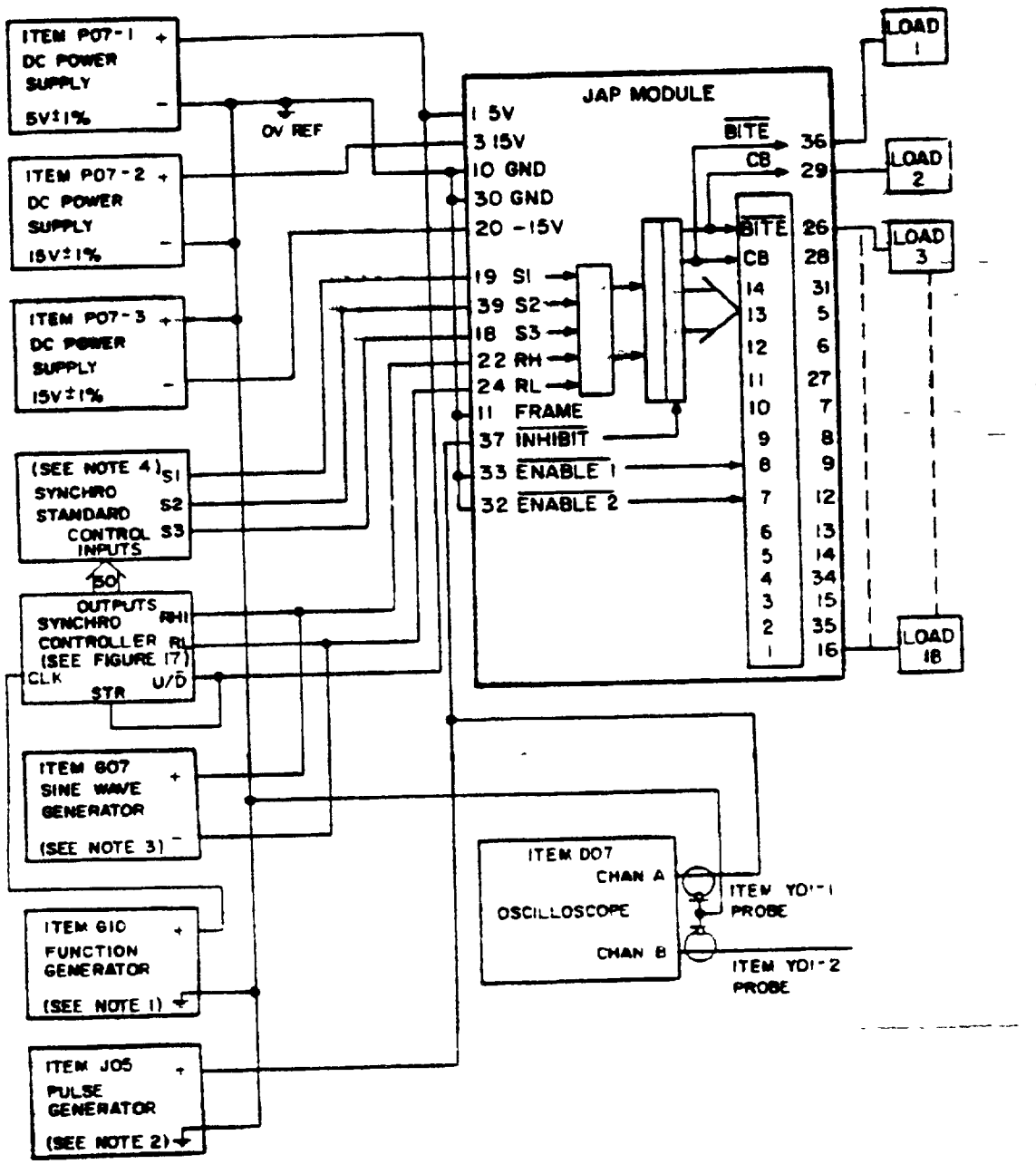


FIGURE 25 Continuous conversion test circuit for table XXIV

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NOTES:
1. Item G10 setup:
   PW = 3.5 V ±10%.
   PW = 27 µs ±10%.
   PRR = 18 kHz ±10%.
   $t_{TLH} = t_{THL} \leq 100$ ns.
2. Item J05 setup:
   PA = 3.5 V ±10%.
   PW = 166.7 µs ±10%.
   $t_{TLH} = t_{THL} \leq 100$ ns.
3. Item G07 setup:
   115 V rms ±10% at 400 Hz ±10%.
4. Synchro standard is ILC Data Device Corporation model SR-460 or equivalent. The output voltage shall be set to 90 V rms L-L position.
5. Loads 1 thru 18 are as shown on figure 4.
6. Monitor pin 29 (CB). The module passes the continuous conversion test if the converter busy pulses continue when item J05 is in the low level state.

FIGURE 25 Continuous conversion test circuit for table XXIV - Continued.

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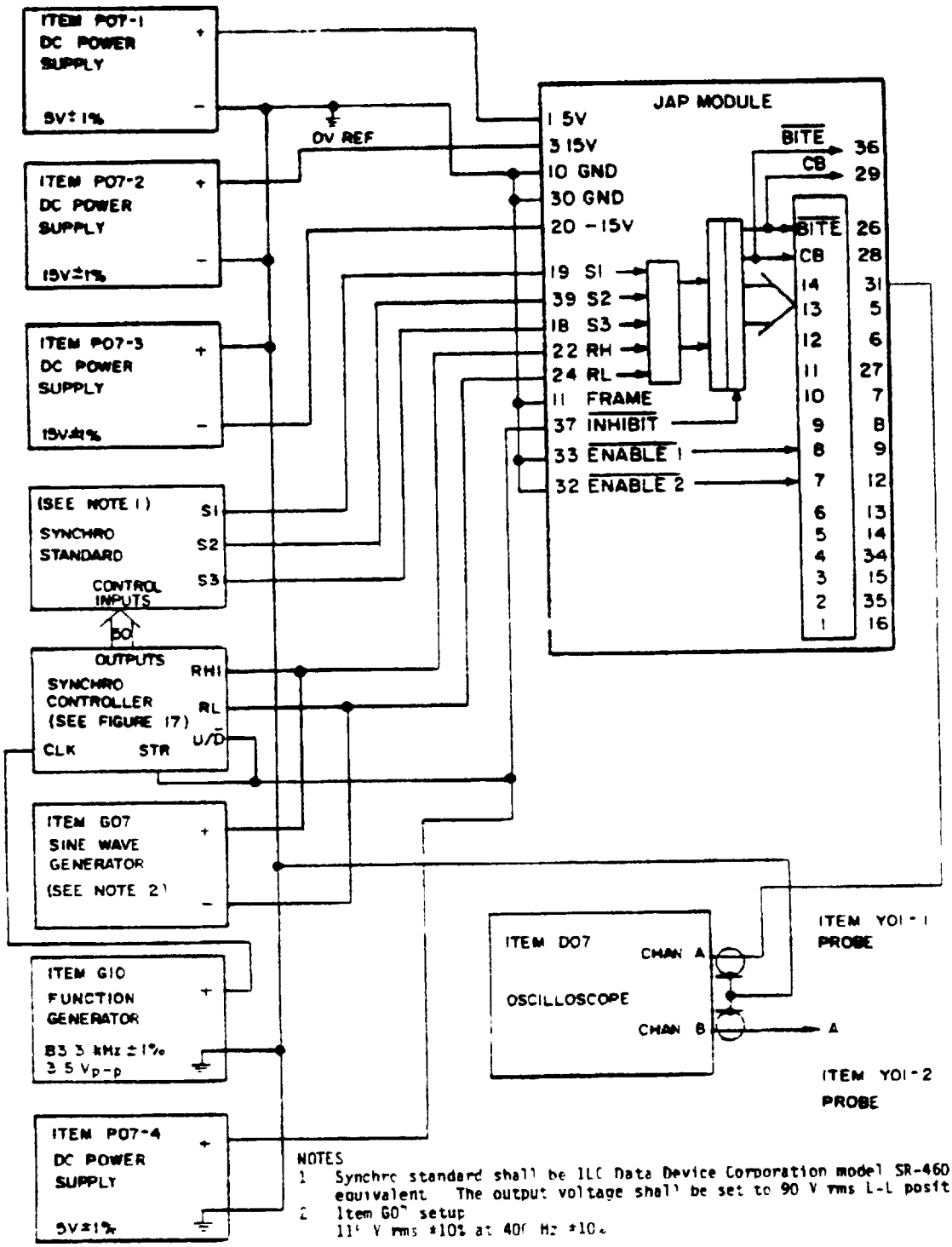
TABLE XXV. Power supply transient amplitude test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
BB1	Power supply transient amplitude ($V_{P_{pk}}$)	1	0	700	mV pk	26,27
BB2	Power supply transient amplitude ($V_{N_{pk}}$)	1	0	700	mV pk	26,27

TABLE XXVI Stress test.

Test condition	Parameter		Initial limits 25°C ±5°C ambient		Unit	Test figure
			Min	Max		
CP	Stress	Following this test the module shall conform to the requirements of all test conditions in table I marked test type X.	--	---	---	28

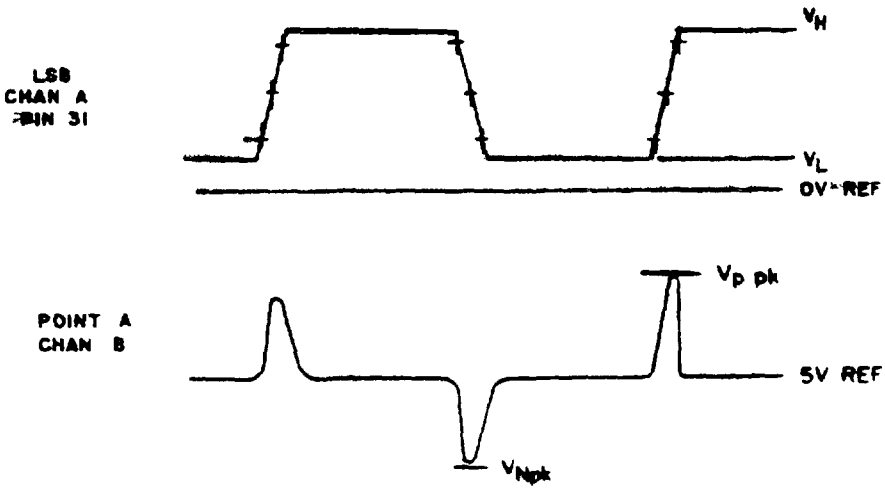
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NOTES
1. Synchro standard shall be ILC Data Device Corporation model SR-460 or equivalent. The output voltage shall be set to 90 V rms L-L position
2. Item G07 setup
   115 V rms ±10% at 400 Hz ±10%

FIGURE 26. Power supply transient amplitude test circuit for table XXV

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NOTES.
1. V_{Ppk} is the most positive peak.
2. V_{Npk} is the most negative peak.
3. Trigger oscilloscope from channel A.

FIGURE 27. Power supply transient amplitude test circuit waveforms for table XXV

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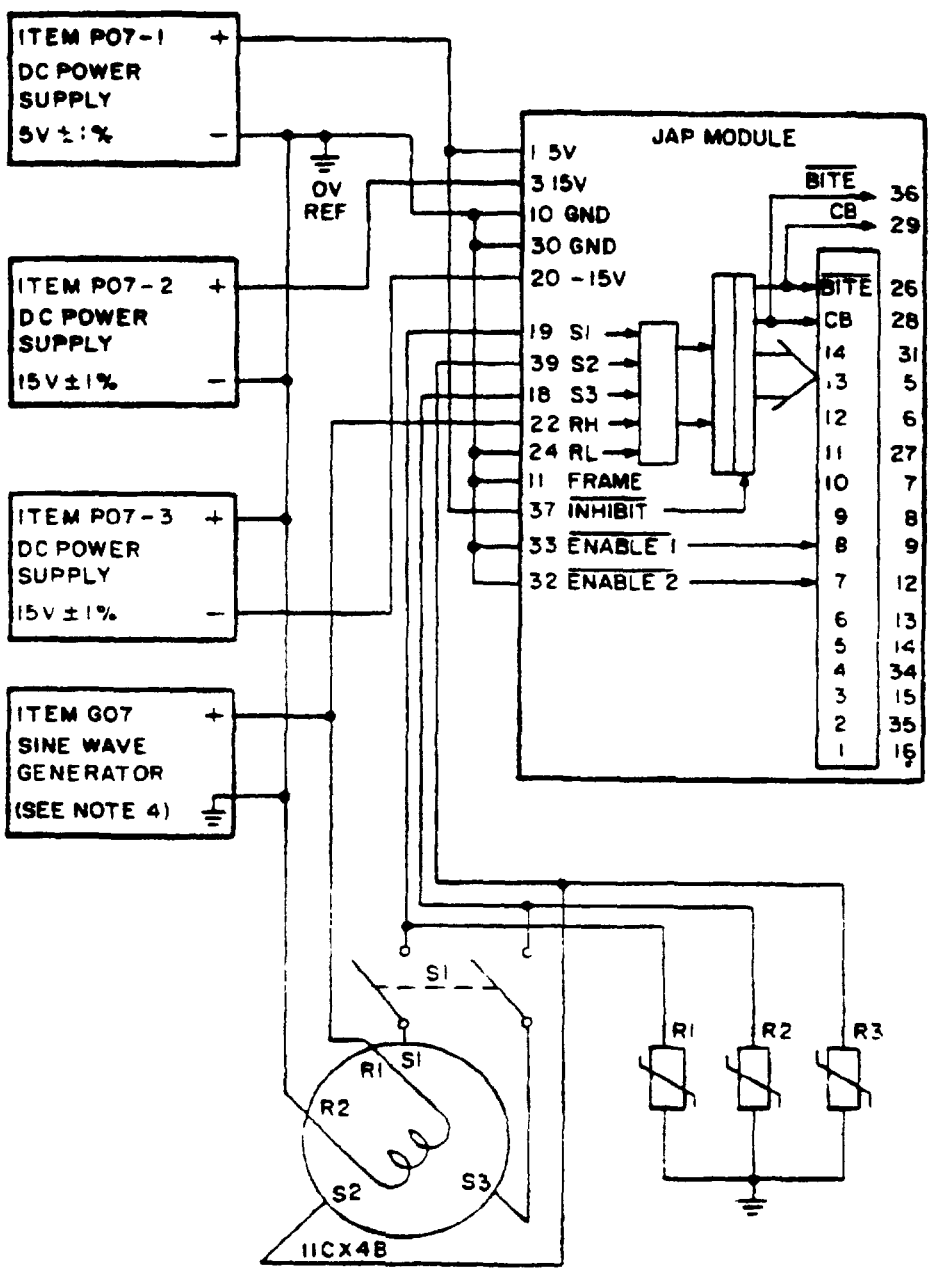


FIGURE 28 Stress test circuit for table XXVI

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NOTES.
1. R1-R3 = V275 LA15A varistor.
2. S1 = DPST switch.
3. Set switch S1 to the closed position for a period of 10 to 30 seconds then operate switch S1 to the open position.
4. Item G07 setup
   115 V rms ±10% at 400 Hz ±10%.
5. This test shall not exceed 1 minute.

FIGURE 28. Stress test circuit for table XXVI - Continued

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